

Minimax Mean Squared Error for Low-Order Network Interference under Bernoulli Assignment

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Abstract

This paper studies design-based estimation of the finite-population all-treated-versus-all-control effect under known bounded-degree interference and low-order polynomial potential-outcome schedules. Under common-probability Bernoulli assignment, coefficient mass bounded by B , graph in- and out-degree bounded by d , and fixed interaction order β , the minimax mean squared error is characterized up to constants depending only on (β, p) :

$$B^2 \min \left\{ 1, \frac{d(k_\star(d, \beta, p))^d}{n} \right\}.$$

Here $k_\star(d, \beta, p)$ is the maximum $r \in \{1, \dots, \min\{\beta, d\}\}$ whose Bernoulli all-treated-versus-all-control contrast is nonzero when this set is nonempty, with $k_\star = 0$ under the degree-zero convention. For each fixed (β, p) , the implied constants are uniform in n , d , and B . The upper bound is achieved by the clipped SNIPE estimator, and the lower bound is obtained from complete-block least-favourable priors built from the block representer. The same block energy A_d governs a uniformly bounded-outcome model class, yielding the equivalent frontier $B^2 \min\{1, dA_d/n\}$. On complete-block graph sequences, the local linear benchmark has exact finite-sample value $B^2 A_d/m$, and asymptotic optimality is characterized by a blockwise quadratic excess criterion. For fair-coin assignment, only odd interaction orders contribute to A_d ; for first-order interactions ($\beta = 1$), the frontier becomes $B^2 \min\{1, d^2/n\}$.

1 Introduction

Network experiments routinely target contrasts between global treatment policies while observing outcomes under randomized assignments that expose only a small portion of the potential-outcome schedule. When one unit’s outcome may depend on nearby units’ assignments, the variance of a graph-aware estimator is shaped both by the size of each exposure neighborhood and by the number of other outcomes that share the same treatment coordinates. This paper gives a finite-population, design-based minimax calculation for that tradeoff under known bounded-degree interference and low-order polynomial response schedules.

The estimand is the average all-treated-versus-all-control contrast. The population, graph, and potential outcomes are fixed, and randomness comes from independent Bernoulli assignment with common treatment probability $p \in (0, 1)$. Each unit’s outcome is a polynomial in the assignments in its graph neighborhood, with interaction order bounded by a fixed β and

*Formal verification details are summarized in the verification appendix.

coefficient mass bounded by B . The graph has maximum in-degree and out-degree at most d , with self-loops counted. This is the coefficient-mass model class in [definition 5](#), and risk is the design mean squared error in [definition 8](#).

The main result, [theorem 1](#), establishes the minimax frontier

$$R_n^*(d, \beta, p, B) \asymp_{\beta, p} B^2 \min \left\{ 1, \frac{d \binom{d}{k_*(d, \beta, p)}}{n} \right\}.$$

The index $k_*(d, \beta, p)$, defined in [definition 1](#), is the maximum $r \in \{1, \dots, \min\{\beta, d\}\}$ whose Bernoulli all-treated-versus-all-control contrast is nonzero when this set is nonempty, with $k_* = 0$ under the degree-zero convention. Thus the rate separates three components: the sample-size factor $1/n$, the graph-overlap factor d , and the number of exposed interaction subsets inside a degree- d neighborhood. The clipped SNIPE estimator in [definition 7](#) attains the upper bound, while the unprojected SNIPE estimator has worst-case mean squared error bounded by the corresponding untruncated variance scale.

The proof identifies two finite-design objects behind the rate. The first is the complete-block score energy A_d introduced in [definition 9](#). [Lemma 1](#) proves that, for complete-block product Bernoulli designs on $\{0, 1\}^d$ and the low-order polynomial space $\mathcal{P}_d$, the block score represents the all-treated-versus-all-control contrast, A_d is the minimum energy of an unbiased local weight, and A_d^{-1} is the minimum energy of a normalized local perturbation. The second is the overlap count in [theorem 1](#), which shows that each r -way treatment subset in one neighborhood can be reused by at most d outcome scores. Together these facts convert local Bernoulli information into the global variance scale dA_d/n , and [lemma 1](#) relates A_d to $\binom{d}{k_*(d, \beta, p)}$.

The lower bound uses complete directed blocks as least-favourable carriers. Within each block, the perturbation is aligned with the normalized Riesz representer h_d , and random baseline coefficients are drawn from a smooth compactly supported density. The two signed priors shift the target by $\rho\delta$ and $-\rho\delta$, while their squared Hellinger distance is controlled by $m\delta^2/(B^2A_d)$. This construction yields the lower frontier at the same scale as the SNIPE upper bound.

The complete-block calculation also gives an exact benchmark for local linear estimators. In [theorem 2](#), for a fixed complete-block graph sequence with block size d_t , $m_t = n_t/d_t$ blocks, and the moment restrictions in [definition 18](#), the local linear minimax risk is

$$R_t^{\text{loc, lin}} = \frac{B^2 A_{d_t}}{m_t}.$$

The theorem gives a finite-sample blockwise quadratic criterion $\Psi_{b,t}$, proves that its excess over $d_t^2 A_{d_t}$ exactly characterizes asymptotic attainment of the benchmark, and shows that risk-optimal local linear weights can remain separated from the canonical complete-block SNIPE score in normalized L^2 distance when $d_t \rightarrow \infty$.

The coefficient-mass model has a bounded-outcome analogue in [definition 21](#). For this class, [theorem 3](#) characterizes the minimax order through the same energy scale:

$$R_{n, \infty}^*(d, \beta, p, B) \asymp_{\beta, p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

The projected estimator is clipped to the natural total-effect range $[-2B, 2B]$, and the unprojected SNIPE estimator remains unbiased with worst-case mean squared error at most $B^2 dA_d/n$. On complete blocks with $d \mid n$, canonical SNIPE has exact worst-case risk $B^2 A_d/m$ for both the coefficient-mass and bounded-outcome classes.

The fair-coin specialization in [theorem 4](#) makes the exposure calculation explicit. At $p = 1/2$, even-order Bernoulli contrasts cancel and odd orders determine the energy:

$$A_d = 4 \sum_{\substack{1 \leq r \leq \bar{\beta}_d \\ r \text{ odd}}} \binom{d}{r}.$$

For first-order interactions ($\beta = 1$), $A_d = 4d$, so the minimax frontier is of order $B^2 \min\{1, d^2/n\}$. This case gives a transparent calibration of the quadratic degree cost induced by first-order interference under fair Bernoulli assignment.

The results build on the design-based potential-outcomes tradition for randomized experiments [[Rubin, 1978](#), [Horvitz and Thompson, 1952](#), [Imbens and Rubin, 2015](#)] and on interference frameworks based on exposure mappings, local dependence, and network randomization [[Sobel, 2006](#), [Hudgens and Halloran, 2008](#), [Aronow and Samii, 2017](#), [Ugander et al., 2013](#), [Eckles et al., 2017](#), [Leung, 2022](#), [Sävje et al., 2021](#), [Hu et al., 2022](#), [Gao and Ding, 2025](#)]. The closest antecedent is the SNIPE analysis for low-order neighborhood interference in [Cortez-Rodriguez et al. \[2023\]](#), which supplies the known-graph Bernoulli estimator and its overlap-aware design variance calculation. The present paper characterizes the matched minimax mean-squared-error frontier in that setting, with explicit degree dependence $dA_d/n \asymp_{\beta,p} d\binom{d}{k_\star}/n$. Its lower bound is generated by a complete-block least-favourable family aligned with the Riesz representer, and its extensions give the bounded-outcome frontier and the exact complete-block local linear benchmark. The calculation also supplies a unit-randomized reference point for recent work on low-order, clustered, and covariate-adjusted interference estimators [[Harshaw et al., 2022](#), [Eichhorn et al., 2024](#), [Wang and Li, 2026](#)]. A Lean 4 development accompanies the paper; the verification appendix states the precise formal scope.

2 Setup and Assumptions

We work in a finite-population design-based setting. The population is the finite unit set V_n with size n ; indices such as i and j refer to units in V_n . A known directed graph G_n records interference, with N_i denoting the assignment coordinates that can enter unit i 's potential outcome. For an assignment vector z , the potential outcome schedule is $Y_i(z)$, and the realized assignment is Z . The observed outcome is $Y_i^{\text{obs}} = Y_i(Z)$. The estimand is the average all-treated-versus-all-control contrast τ_n . This finite-population design-based randomization world and the accompanying fixed-potential-outcome polynomial interference world provide the objects used in all risk statements below.

The assignment mechanism is the independent Bernoulli design. We write p for the common treatment probability, P_Z for the product assignment law, and $\mathbb{E}_Z$ for expectation under that law.

The notation is grouped by role in the results.

Object group	Role	First main use
$(V_n, G_n, N_i, Z, Y_i^{\text{obs}}, \tau_n), c, (d, \beta, B, p)$	finite-population decision problem	theorem 1
$g_{i,\beta,p}, \hat{\tau}_n^{\text{SNIPE}}, \hat{\tau}_n^{\text{up}}, R_n^*$	score, estimator, and coefficient-class risk	theorem 1
$g_d, A_d, h_d, L^2(P_Z), \mathcal{P}_d, L_d$	complete-block quantities	Riesz theorems 1 and 2
$V_b, U_b, \delta, \rho, \Pi_+, \Pi_-$	least-favourable construction	block theorem 1
$G_t, n_t, d_t, m_t, b(i), w_{i,t}, \mathcal{E}_t^{\text{loc,lin}}, \Psi_{b,t}$	complete-block local linear benchmark	theorem 2
$\mathcal{M}_{n,d,\beta}^\infty(B), R_{n,\infty}^*, \hat{\tau}_{n,\infty}^{\text{up}}$	uniformly bounded-outcome comparison	theorem 3

The setup first introduces the design, graph, model class, estimator, and risk. The block programs, least-favourable laws, local-linear sequence, and bounded-outcome envelope then appear at the points where the main theorems use them.

Assumption 1. `[ass:bernoulli-design]` (Bernoulli assignment) *The assignment law P_Z is the common-probability product Bernoulli design with $p \in (0, 1)$. Equivalently, the assignments $\{Z_j : j \in V_n\}$ are mutually independent and*

$$\Pr(Z_j = 1) = p$$

for every $j \in V_n$.

[Assumption 1](#) is the standard unit-level Bernoulli randomization condition used in design-based analyses of SNIPE-type estimators [[Cortez-Rodriguez et al., 2023](#)]. Its independence and common-probability clauses make the finite-design moments of centered assignment monomials explicit.

Interference is local in the known graph. The parameter d bounds both the number of assignment coordinates entering each unit’s outcome and the number of outcomes affected by each assignment coordinate, with self-loops counted in both directions.

Assumption 2. `[ass:bounded-degree]` (Bounded interference degree) *Self-loops are allowed and count in both degrees. The interference neighborhoods satisfy*

$$\max_{i \in V_n} |N_i| \leq d \quad \text{and} \quad \max_{j \in V_n} |\{i \in V_n : j \in N_i\}| \leq d.$$

[Assumption 2](#) is the standard bounded in- and out-degree neighborhood interference condition with self-loops [[Cortez-Rodriguez et al., 2023](#)]. The first bound limits the dimension of each unit’s exposure polynomial, while the second controls how many summands in an estimator can share a treatment coordinate.

Potential outcomes are represented by raw assignment monomials on the relevant neighborhood. The coefficient schedule is c , with $c_{i,S}$ denoting the coefficient on a finite subset S of assignment coordinates. The interaction order β fixes the maximum monomial degree used by the model.

Assumption 3. `[ass:low-order]` (Low-order interactions) *For every $i \in V_n$ and every finite subset $S \subseteq V_n$,*

$$c_{i,S} = 0$$

whenever $|S| > \beta$.

[Assumption 3](#) is the standard bounded polynomial interaction order condition [[Cortez-Rodriguez et al., 2023](#)]. It turns the neighborhood outcome schedule into a low-order polynomial and separates the role of graph degree from the role of interaction order.

The coefficient envelope B normalizes the size of each unit's outcome polynomial.

Assumption 4. `[ass:bounded-coefficient-mass]` (Bounded coefficient mass) *For every $i \in V_n$,*

$$\sum_{S \subseteq N_i} |c_{i,S}| \leq B.$$

[Assumption 4](#) is the standard SNIPE $Y_{\max}$ coefficient-mass normalization [[Cortez-Rodriguez et al., 2023](#)]. It supplies a uniform scale for the total-effect target and for the design mean squared error comparisons.

The exposed interaction order records how much of the polynomial class can be seen within a degree- d neighborhood. The Bernoulli contrast coefficient is $\Delta_r(p)$, with r indexing interaction order; it is the order- r all-treated-versus-all-control contrast of a centered assignment monomial,

$$\Delta_r(p) := (1 - p)^r - (-p)^r.$$

Whether $\Delta_r(p)$ vanishes decides which interaction orders the design can expose, and hence the value of $k_\star$ below.

Definition 1. `[def:exposed-order]` (Exposed order $\bar{\beta}_d$ and $k_\star(d, \beta, p)$) *For every $d \geq 0$, set*

$$\bar{\beta}_d := \min\{\beta, d\}.$$

In particular, $\bar{\beta}_0 = 0$.

If $\bar{\beta}_d \geq 1$, define

$$k_\star(d, \beta, p) := \max\{r \in \{1, \dots, \bar{\beta}_d\} : \Delta_r(p) \neq 0\}.$$

Since $\Delta_1(p) = 1$, the index set contains $r = 1$, so this maximum is well defined.

If $\bar{\beta}_d = 0$, equivalently $d = 0$ or $\beta = 0$, set

$$k_\star(d, \beta, p) := 0$$

as a totality convention rather than as an attained interaction order. Thus the maximum case applies exactly when $d \geq 1$ and $\beta \geq 1$. Throughout the subsequent results, $\beta \geq 1$ is fixed.

The quantities $\bar{\beta}_d$ and $k_\star(d, \beta, p)$ determine which centered monomial contrasts carry identifying content under the Bernoulli design.

Definition 2. `[def:zero-degree-conventions]` (Zero-degree conventions $\binom{0}{0}$, g_0 , A_0 , and h_0) *For every $\beta \in \mathbb{N}$ and $p \in \mathbb{R}$, the degree-zero conventions are*

$$\bar{\beta}_0 := \min\{\beta, 0\} = 0, \quad k_\star(0, \beta, p) := 0, \quad g_0 := 0, \quad A_0 := 0.$$

The normalized representer is guarded at degree zero by setting

$$h_0 := 0, \quad h_{0,T} := 0$$

for every block-coordinate subset T of the zero-coordinate block, with h_0 imposed as an empty-object convention rather than as the quotient g_0/A_0 .

These conventions allow the formulas for scores, energies, and representers to remain total across degree values, while the main results use the positive-degree regime stated in their conditions.

We next collect the graph and coefficient classes. The graph carrier is $\mathcal{G}_{n,d}$, the coefficient carrier is $\mathcal{C}_{n,d,\beta}(B)$, and their product model class is $\mathcal{M}_{n,d,\beta}(B)$.

Definition 3. [def:graph-class] (Graph class $\mathcal{G}_{n,d}$) *The graph class with degree bound d is*

$$\mathcal{G}_{n,d} := \{G_n \subseteq V_n \times V_n : G_n \text{ satisfies } \text{assumption 2}\}.$$

Definition 4. [def:coefficient-class] (Coefficient class $\mathcal{C}_{n,d,\beta}(B)$) *The coefficient class with interaction order β and mass radius B is*

$$\mathcal{C}_{n,d,\beta}(B) := \{c = (c_{i,S})_{i \in V_n, S \subseteq V_n} : c_{i,S} = 0 \text{ whenever } S \not\subseteq N_i, c \text{ satisfies } \text{assumptions 3 and 4}\}.$$

Definition 5. [def:model-class] (Model class $\mathcal{M}_{n,d,\beta}(B)$) *Under assumptions 2 to 4, with coefficient schedules supported on the in-neighborhoods in the sense that $c_{i,S} = 0$ whenever $S \not\subseteq N_i$, the model class is*

$$\mathcal{M}_{n,d,\beta}(B) := \{(G_n, c) : G_n \in \mathcal{G}_{n,d}, c \in \mathcal{C}_{n,d,\beta}(B)\}.$$

Together, definitions 3 to 5 specify the coefficient-mass model over which the first minimax risk is evaluated.

The estimator uses centered Bernoulli contrast scores. For unit i , the score $g_{i,\beta,p}$ weights observed outcomes by all centered neighborhood monomials up to the exposed order.

Definition 6. [def:snipe-score] (SNIPE score $g_{i,\beta,p}$) *For unit i , the SNIPE score is*

$$g_{i,\beta,p}(z) := \sum_{r=1}^{\bar{\beta}_d} \frac{\Delta_r(p)}{[p(1-p)]^r} \sum_{\substack{S \subseteq N_i \\ |S|=r}} \prod_{j \in S} (z_j - p).$$

The unprojected estimator $\hat{\tau}_n^{\text{SNIPE}}$ averages observed outcomes weighted by these scores, and the projected estimator $\hat{\tau}_n^{\text{up}}$ clips that average to $[-B, B]$ for the coefficient-mass risk comparison.

Definition 7. [def:snipe-estimator] (SNIPE estimator $\hat{\tau}_n^{\text{SNIPE}}$) *The unprojected SNIPE estimator is*

$$\hat{\tau}_n^{\text{SNIPE}} := \frac{1}{n} \sum_{i \in V_n} Y_i^{\text{obs}} g_{i,\beta,p}(Z).$$

Its projected version is

$$\hat{\tau}_n^{\text{up}} := \Pi_{[-B,B]}(\hat{\tau}_n^{\text{SNIPE}}),$$

where $\Pi_{[-B,B]}$ denotes Euclidean projection onto $[-B, B]$.

The decision problem is graph-aware and design-based: estimators may use the graph, the assignment, and the observed outcomes, while risk averages only over the random assignment. The generic estimator is $\hat{\tau}_n$, and the resulting coefficient-class minimax risk is $R_n^*(d, \beta, p, B)$. This is the design-based decision-theoretic world used by the main theorem.

Definition 8. `[def:minimax-risk]` (Minimax risk $R_n^*(d, \beta, p, B)$) The minimax design mean squared error over $\mathcal{M}_{n,d,\beta}(B)$ is

$$R_n^*(d, \beta, p, B) := \inf_{\hat{\tau}_n} \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right],$$

where the infimum ranges over estimators $\hat{\tau}_n$ measurable with respect to $(G_n, Z, \{Y_i^{\text{obs}} : i \in V_n\})$.

The lower-bound construction and the local linear benchmark both reduce to complete blocks. For one d -block, the block score g_d , energy A_d , normalized representer h_d , block-coordinate subset T , and raw coefficient $h_{d,T}$ are as follows.

Definition 9. `[def:block-score-energy]` (Complete-block score and energy g_d, A_d, h_d) For $d \geq 1$ and a Bernoulli assignment vector $z = (z_1, \dots, z_d)$, define

$$g_d(z) := \sum_{r=1}^{\bar{\beta}_d} \frac{\Delta_r(p)}{[p(1-p)]^r} \sum_{\substack{S \subseteq \{1, \dots, d\} \\ |S|=r}} \prod_{j \in S} (z_j - p).$$

The corresponding energy is

$$A_d := \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r}.$$

The normalized block representer is

$$h_d := \frac{g_d}{A_d}.$$

Its raw-monomial expansion is

$$h_d(z) = \sum_{T \subseteq \{1, \dots, d\}} h_{d,T} \prod_{j \in T} z_j.$$

The least-favourable family places complete directed blocks inside the population. Its notation includes the number of blocks m , active units q , active fraction ρ , active block V_b , block index b , baseline halfwidth s , density f_s , block baseline U_b , representer mass bound $H_{\beta,p}$, perturbation amplitude δ , sign σ , and prior Π_σ . This is the least-favourable block world used to calibrate the minimax lower bound.

Definition 10. `[def:block-family]` (Least-favourable block family Π_σ) For $B > 0$ and $1 \leq d \leq n$, set

$$m := \lfloor n/d \rfloor, \quad q := md, \quad \rho := q/n.$$

Partition q units into m blocks V_b of size d . Each block contains every within-block directed arrow, including loops, and the remaining $n - q$ units are isolated.

Set

$$s := B/2, \quad f_s(u) := s^{-1} \cos^2\left(\frac{\pi u}{2s}\right) \mathbf{1}\{|u| \leq s\}.$$

Let $U_1, \dots, U_m$ be independent draws from density f_s . Define

$$H_{\beta,p} := \sup_{d \geq 1} \sum_T |h_{d,T}|$$

and choose

$$\delta := B \min\{(2H_{\beta,p})^{-1}, (4\pi)^{-1}\} \min\{1, \sqrt{A_d/m}\}.$$

Under Π_σ , each active unit $i \in V_b$ has coefficients

$$c_{i,T} := U_b \mathbf{1}\{T = \emptyset\} + \sigma \delta h_{d,T}, \quad |T| \leq k_\star.$$

All remaining coefficients, including coefficients of inactive units, are zero. The variables U_b randomize fixed potential-outcome schedules as prior draws over schedules.

The two signed versions of the construction also induce prior-predictive laws. The negative law is named separately because the affinity comparison in the lower-bound argument uses the pair symmetrically.

Definition 11. [synth_4] (Negative least-favourable prior-predictive law Π_-) We write Π_- for the prior-predictive law induced by the least-favourable block construction with sign $\sigma = -1$: draw $Z \sim \prod_{j \in V_n} \text{Bernoulli}(p)$ and independent $U_1, \dots, U_m$ with density f_s , set the active coefficients in block V_b to $c_{i,T} = U_b \mathbf{1}\{T = \emptyset\} - \delta h_{d,T}$ for $|T| \leq k_\star$ and all remaining coefficients to zero, and take the induced joint law of $(Z, \{Y_i^{\text{obs}} : i \in V_n\})$.

Definition 12. [synth_3] (Positive score-aligned block prior Π_+) In the complete-block construction with $B > 0$ and $1 \leq d \leq n$, we write $\Pi_+(n, d, \beta, B, p)$ for the probability law on fixed coefficient schedules induced by independent draws $U_b \sim f_s$ under which each active unit $i \in V_b$ has $c_{i,T} = U_b \mathbf{1}\{T = \emptyset\} + \delta h_{d,T}$ for $|T| \leq k_\star(d, \beta, p)$, and every other coefficient, including each coefficient of an inactive unit, equals 0.

For the complete-block Riesz calculations, the finite design space and the polynomial contrast space are useful in their own right. We use the block-local Riesz world: the block polynomial subspace $\mathcal{P}_d$, candidate polynomial f , candidate perturbation h , all-one-versus-all-zero contrast functional L_d , candidate block-local weight w , and the associated minimum-energy programs. The contrast functional evaluates a block polynomial at the all-treated and all-control assignments,

$$L_d(f) := f(\mathbf{1}_d) - f(\mathbf{0}_d),$$

so the normalization $L_d(h) = 1$ below fixes the scale at which a perturbation moves the target. Writing A_i for the local SNIPE energy of unit i , the same construction applied to that unit's own neighborhood gives

$$A_i := \sum_{r=1}^{\min\{\beta, |N_i|\}} \binom{|N_i|}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r},$$

which reduces to the complete-block energy A_d when $|N_i| = d$.

Definition 13. [synth_2] (Design $L^2(P_Z)$ space) We write $L^2(P_Z)$ for the real vector space of functions $w : \{0, 1\}^{\{1, \dots, d\}} \rightarrow \mathbb{R}$ with finite design second moment $\mathbb{E}_Z[w(Z)^2] < \infty$ under the product Bernoulli assignment law P_Z with success probability p .

Definition 14. [synth_1] (Block polynomial space $\mathcal{P}_d$) For $d \geq 1$, $\mathcal{P}_d$ is the linear subspace of $L^2(P_Z)$ spanned by the raw monomials $z \mapsto \prod_{j \in S} z_j$ with $S \subseteq \{1, \dots, d\}$ and $|S| \leq \bar{\beta}_d$.

Definition 15. `[def:perturbation-program]` (Perturbation program $\text{PerturbProg}_{d,\beta,p}$) For $d \geq 1$, define

$$\text{PerturbProg}_{d,\beta,p} := \min_{h \in \mathcal{P}_d} \mathbb{E}_Z[h(Z)^2]$$

subject to

$$L_d(h) = 1.$$

Definition 16. `[def:weight-program]` (Weight program $\text{WeightProg}_{d,\beta,p}$) For $d \geq 1$, define

$$\text{WeightProg}_{d,\beta,p} := \min_{w \in L^2(P_Z)} \mathbb{E}_Z[w(Z)^2]$$

subject to

$$\mathbb{E}_Z[w(Z)] = 0$$

and, for every nonempty $S \subseteq \{1, \dots, d\}$ with $|S| \leq \bar{\beta}_d$,

$$\mathbb{E}_Z \left[w(Z) \prod_{j \in S} Z_j \right] = 1.$$

The perturbation program $\text{PerturbProg}_{d,\beta,p}$ and the weight program $\text{WeightProg}_{d,\beta,p}$ express the same complete-block information through dual normalizations: one fixes the all-one-versus-all-zero contrast and minimizes polynomial energy, while the other fixes unbiased raw-monomial moments and minimizes weight energy.

The fixed-graph local linear benchmark specializes the population sequence to complete blocks. Let G_t be the fixed complete-block graph at index t , with carrier V_{n_t} , size n_t , block size d_t , and neighborhoods $N_i^{G_t}$. The local coefficient class is $\mathcal{M}_t(G_t)$.

Definition 17. `[def:local-linear-class]` (Fixed-graph class $\mathcal{M}_t(G_t)$) Under [assumptions 3](#) and [4](#), on the fixed graph G_t , define $\mathcal{M}_t(G_t)$ as the set of coefficient schedules c satisfying

$$c_{i,S} = 0$$

whenever $S \subseteq N_i^{G_t}$ and $|S| \leq \bar{\beta}_{d_t}$ fail to hold, and satisfying, for every $i \in V_{n_t}$,

$$\sum_{S \subseteq N_i^{G_t}} |c_{i,S}| \leq B.$$

A local linear estimator uses a block-local assignment weight $w_{i,t}$. If $b(i)$ denotes the block containing unit i , then $Z_{b(i)}$ is the assignment vector in that block, $\hat{\tau}_{w,t}$ is the resulting estimator, and $\mathcal{E}_t^{\text{loc},\text{lin}}$ is the moment-restricted class.

Definition 18. `[def:local-linear-estimator-class]` (Local linear estimators $\mathcal{E}_t^{\text{loc},\text{lin}}$) Define $\mathcal{E}_t^{\text{loc},\text{lin}}$ as the set of estimators

$$\hat{\tau}_{w,t} := n_t^{-1} \sum_{i \in V_{n_t}} w_{i,t}(Z_{b(i)}) Y_i(Z)$$

such that $w_{i,t} \in L^2(P_Z)$ is measurable with respect to $Z_{b(i)}$,

$$\mathbb{E}_Z[w_{i,t}] = 0,$$

and, for every nonempty $S \subseteq N_i^{G_t}$ with $|S| \leq \bar{\beta}_{d_t}$,

$$\mathbb{E}_Z \left[w_{i,t} \prod_{j \in S} Z_j \right] = 1.$$

The fixed-graph local linear risk $R_t^{\text{loc,lin}}$ evaluates these estimators over the corresponding coefficient schedules.

Definition 19. `[def:local-linear-risk]` (Local linear risk $R_t^{\text{loc,lin}}$) For a fixed complete-block graph G_t , define

$$R_t^{\text{loc,lin}} := \inf_{\hat{\tau}_{w,t} \in \mathcal{E}_t^{\text{loc,lin}}} \sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z \left[(\hat{\tau}_{w,t} - \tau_{n_t})^2 \right].$$

The supremum ranges over coefficient schedules on the fixed graph G_t .

Finally, we introduce the uniformly bounded-outcome class. Its schedule carrier is $\mathcal{C}_{n,d,\beta}^\infty(B)$, and its model class is $\mathcal{M}_{n,d,\beta}^\infty(B)$.

Definition 20. `[def:bounded-outcome-coefficient-class]` (Bounded-outcome coefficient class $\mathcal{C}_{n,d,\beta}^\infty(B)$) Under [assumption 3](#), define

$$\mathcal{C}_{n,d,\beta}^\infty(B) := \left\{ c : c_{i,S} = 0 \text{ whenever } S \not\subseteq N_i, \quad c_{i,S} = 0 \text{ whenever } |S| > \beta, \quad \sup_{z \in \{0,1\}^{V_n}} |Y_i(z)| \leq B \text{ for every } i \in V_n \right\}$$

Definition 21. `[def:bounded-outcome-model-class]` (Bounded-outcome model class $\mathcal{M}_{n,d,\beta}^\infty(B)$) Under [assumptions 2 and 3](#), define

$$\mathcal{M}_{n,d,\beta}^\infty(B) := \{ (G_n, c) : G_n \in \mathcal{G}_{n,d}, \quad c \in \mathcal{C}_{n,d,\beta}^\infty(B) \}.$$

The notation for the two minimax risks distinguishes the coefficient-mass class from the uniformly bounded-outcome class. We write $R_{n,\ell_1}^*(d, \beta, p, B)$ for the coefficient-mass risk and $R_{n,\infty}^*(d, \beta, p, B)$ for the bounded-outcome risk.

Definition 22. `[def:two-class-minimax-risks]` (Two minimax risks R_{n,ℓ_1}^* and $R_{n,\infty}^*$) For $0 \leq p \leq 1$, define

$$R_{n,\ell_1}^*(d, \beta, p, B) := R_n^*(d, \beta, p, B)$$

and

$$R_{n,\infty}^*(d, \beta, p, B) := \inf_{\hat{\tau}_n} \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right].$$

The infimum ranges over graph-aware estimators $\hat{\tau}_n$ of G_n , Z , and $\{Y_i^{\text{obs}} : i \in V_n\}$ that, for each fixed G_n and Z , are measurable as functions of the observed-outcome vector and whose modelwise bounded-outcome risk range

$$\left\{ \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right] : (G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B) \right\}$$

is bounded above.

For the bounded-outcome comparison, the SNIPE estimator is projected to the natural range of the total-effect contrast. The clipped version is $\hat{\tau}_{n,\infty}^{\text{up}}$.

Definition 23. [def:bounded-outcome-clipped-snipe] (Bounded-outcome clipped SNIPE $\hat{\tau}_{n,\infty}^{\text{up}}$) Define

$$\hat{\tau}_{n,\infty}^{\text{up}} := \Pi_{[-2B,2B]}(\hat{\tau}_n^{\text{SNIPE}}),$$

where $\Pi_{[-2B,2B]}$ denotes Euclidean projection onto $[-2B, 2B]$.

Two prior laws are compared through the squared Hellinger distance, used throughout in its unhalved form: for laws P and Q with densities f and g with respect to a common dominating measure λ ,

$$H^2(P, Q) := \int \left(\sqrt{f} - \sqrt{g} \right)^2 d\lambda.$$

Every constant reported for a least-favourable pair below refers to this normalization; the halved convention would rescale those constants by a factor of two.

The definitions above fix the design, model classes, estimators, block programs, and risk criteria used in the results that follow.

3 Main Results

The definitions in [definitions 5, 7 and 8](#) lead to a design-based risk comparison in which the graph is known, treatment is assigned independently, and performance is evaluated uniformly over the coefficient-mass class. The first result characterizes the degree dependence of that comparison. Its rate is governed by the exposed Bernoulli contrast order $k_*(d, \beta, p)$, equivalently by the leading nonzero centered monomial order exposed by the assignment design when a positive-order exposed contrast is present and by the degree-zero convention otherwise. The projected SNIPE estimator from [definition 7](#) attains the same order as the lower bound, while the unprojected estimator has the corresponding nontruncated variance bound.

Theorem 1. [thm:degree-frontier] (Degree Frontier Bounds) Fix $\beta \geq 1$ and $p \in (0, 1)$. Under the notation in [definitions 1, 5 and 8](#), there exist constants $c_{\beta,p}$ and $C_{\beta,p}$ such that $0 < c_{\beta,p} \leq C_{\beta,p}$ and, for every $n, d \in \mathbb{N}$ and $B \in \mathbb{R}$ with $n \geq 1$, $d \leq n$, and $B \geq 0$, the following statements hold:

- (Minimax frontier.)

$$c_{\beta,p} B^2 \min \left\{ 1, \frac{d(k_*(d, \beta, p))}{n} \right\} \leq R_n^*(d, \beta, p, B).$$

Moreover,

$$R_n^*(d, \beta, p, B) \leq \sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] \leq C_{\beta,p} B^2 \min \left\{ 1, \frac{d(k_*(d, \beta, p))}{n} \right\}.$$

The unprojected SNIPE estimator also satisfies

$$\sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] \leq C_{\beta,p} B^2 \frac{d(k_*(d, \beta, p))}{n}.$$

- (Score energy.) For every $(G, c) \in \mathcal{M}_{n,d,\beta}(B)$ and every unit $i \in V_n$,

$$A_i \leq A_d.$$

- **(Overlap counts.)** For every $(G, c) \in \mathcal{M}_{n,d,\beta}(B)$, every unit $i \in V_n$, and every integer $r \geq 1$,

$$\sum_{\ell \in V_n} \binom{|N_i \cap N_\ell|}{r} = \sum_{\substack{S \subseteq N_i \\ |S|=r}} \#\{\ell \in V_n : S \subseteq N_\ell\} \leq d \binom{|N_i|}{r} \leq d \binom{d}{r}.$$

- **(Block construction.)** If $B > 0$ and $d \geq 1$, let $m = \lfloor n/d \rfloor$, let ρ be the active population fraction, and let δ be the least-favourable perturbation amplitude. For every block-coefficient vector $U = (U_1, \dots, U_m)$ satisfying $|U_b| \leq B/2$ for all b , there exist two models M_+ and M_- in $\mathcal{M}_{n,d,\beta}(B)$, both supported on the complete d -block graph, whose coefficient schedules are the corresponding $+1$ and -1 block schedules, and whose total treatment effects satisfy

$$\tau_n(M_+) = \rho\delta, \quad \tau_n(M_-) = -\rho\delta.$$

The associated block prior densities satisfy

$$H^2(\Pi_+, \Pi_-) \leq \frac{4\pi^2 m \delta^2}{B^2 A_d}.$$

In addition,

$$\frac{1}{2} \leq \rho \leq 1, \quad \frac{A_d}{m} = \frac{dA_d/n}{\rho}.$$

[Theorem 1](#) identifies the relevant effective dimension as a degree-weighted count of exposed interactions. The upper bound combines unbiasedness of the SNIPE score with the overlap count in the theorem, so the risk grows with the number of units whose scores can share treatment coordinates. The lower bound is calibrated on complete blocks: the perturbation is aligned with the normalized block representer of [definition 9](#), and the Hellinger bound makes the target separation comparable to the same block energy. Thus the minimax comparison measures the exact price of estimating the total treatment effect under known bounded-degree interference and low-order polynomial response schedules.

The next result isolates what local linear estimators can achieve on a fixed complete-block sequence. This benchmark is useful because complete blocks are the least-favourable carriers used above and because local linear estimators are a natural graph-local analogue of inverse-probability weighted estimators in randomized experiments. The characterization is exact at each t , and the asymptotic statements distinguish attainment of the optimal block risk from closeness to the canonical SNIPE score itself.

Theorem 2. [[thm:sharp-local-linear-constant-and-representers](#)] (Sharp Local Linear Representers) Fix $\beta \in \mathbb{N}$, $p \in (0, 1)$, and $B > 0$, with $\beta \geq 1$. Suppose that:

- **(Block sizes.)** For each t , n_t and d_t are positive integers and $d_t \mid n_t$. Write $m_t := n_t/d_t$.
- **(Block graph.)** For each t , G_t is the directed complete-block graph on V_{n_t} : units are partitioned into m_t active blocks V_b of size d_t , each block contains all directed edges including self-loops, and there are no edges between distinct blocks.
- **(Assignment law.)** For each t , the finite design D_t satisfies [assumption 1](#) with common treatment probability p .

Then, for every t ,

$$R_t^{\text{loc,lin}} = \frac{B^2 A_{d_t}}{m_t} = \frac{B^2 d_t A_{d_t}}{n_t}.$$

For every t and every $w \in \mathcal{E}_t^{\text{loc,lin}}$,

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w,t} - \tau_{n_t})^2] = \frac{B^2}{n_t^2} \sum_{b=1}^{m_t} \Psi_{b,t}(w),$$

where, with $Z_S := \prod_{j \in S} Z_j$,

$$\Psi_{b,t}(w) := \max_{\substack{(\varepsilon_i, S_i)_{i \in V_b} \\ \varepsilon_i \in \{-1, 1\}, S_i \subseteq V_b, |S_i| \leq \bar{\beta}_{d_t}}} \mathbb{E}_Z \left[\left(\sum_{i \in V_b} \varepsilon_i \{w_{i,t}(Z_b) Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\} \right)^2 \right].$$

Moreover, for every t , every $w \in \mathcal{E}_t^{\text{loc,lin}}$, and every block V_b ,

$$d_t^2 A_{d_t} \leq \Psi_{b,t}(w).$$

For every sequence $w_t \in \mathcal{E}_t^{\text{loc,lin}}$,

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} \longrightarrow 1$$

if and only if

$$\frac{1}{m_t d_t^2 A_{d_t}} \sum_{b=1}^{m_t} \{ \Psi_{b,t}(w_t) - d_t^2 A_{d_t} \} \longrightarrow 0.$$

For every sequence $w_t \in \mathcal{E}_t^{\text{loc,lin}}$, the condition

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z \left[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2 \right] \longrightarrow 0$$

implies

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} \longrightarrow 1.$$

If $d_t \rightarrow \infty$, then there exists a sequence $w_t \in \mathcal{E}_t^{\text{loc,lin}}$ such that

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} \longrightarrow 1,$$

and, for all sufficiently large t ,

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z \left[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2 \right] = 2.$$

For every sufficiently large t , the same sequence also satisfies, for every bijection $\pi : V_{n_t} \rightarrow V_{n_t}$ that preserves block membership,

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z \left[\left\{ w_{i,t}(Z_{b(i)}) - g_{d_t}((Z_{\pi^{-1}(j)})_{j \in V_{b(i)}}) \right\}^2 \right] = 2.$$

Theorem 2 gives the exact complete-block constant $B^2 A_{d_t}/m_t$. The block criterion $\Psi_{b,t}$ is the finite-dimensional worst-case quadratic form induced by the moment restrictions in [definition 18](#); averaging its excess over $d_t^2 A_{d_t}$ is necessary and sufficient for asymptotic minimaxity within the local linear class. The final clauses show that score closeness is a sufficient route to the benchmark, while the benchmark also admits asymptotically optimal local linear weights whose normalized distance from the canonical complete-block score is fixed at two along growing block sizes. The result therefore separates the risk-optimality property from uniqueness of the SNIPE representer as a weight function.

The coefficient-mass condition in [definition 5](#) has a direct bounded-outcome counterpart in [definition 21](#). The next theorem compares the two decision problems and shows that the same complete-block energy controls the minimax order under the uniformly bounded-outcome envelope. This formulation aligns the SNIPE variance calculation with the design-based bounded-outcome risk scale used in network-experiment analyses such as [Cortez-Rodriguez et al. \[2023\]](#) and related low-order interference work [\[Harshaw et al., 2022, Wang and Li, 2026\]](#).

Theorem 3. `[thm:bounded-outcome-degree-frontier]` (Bounded-Outcome Degree Frontier) *Fix an interaction order $\beta \geq 1$ and a treatment probability $p \in (0, 1)$. There exist real constants $c_{\beta,p}$ and $C_{\beta,p}$ with*

$$0 < c_{\beta,p} \leq C_{\beta,p}$$

such that, for every $n \geq 1$, every design D satisfying [assumption 1](#) with common treatment probability p , every integer $d \leq n$, and every $B \geq 0$, the following statements hold.

- **(Model comparison.)** *The coefficient-mass class $\mathcal{M}_{n,d,\beta}(B)$ of [definition 5](#) is contained, carrier by carrier, in the uniformly bounded-outcome class $\mathcal{M}_{n,d,\beta}^\infty(B)$ of [definition 21](#). If $B > 0$ and $d \geq 1$, this containment is strict.*
- **(Frontier bounds.)** *With A_d denoting the complete-block score energy,*

$$c_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\} \leq R_{n,\ell_1}^*(d, \beta, p, B),$$

$$R_{n,\ell_1}^*(d, \beta, p, B) = R_n^*(d, \beta, p, B),$$

$$R_{n,\ell_1}^*(d, \beta, p, B) \leq R_{n,\infty}^*(d, \beta, p, B),$$

and

$$R_{n,\infty}^*(d, \beta, p, B) \leq \sup_{\mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2 \right] \leq C_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

The coefficient-class clipped SNIPE estimator satisfies

$$\sup_{\mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] \leq C_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

- **(Unclipped SNIPE.)** *On both $\mathcal{M}_{n,d,\beta}(B)$ and $\mathcal{M}_{n,d,\beta}^\infty(B)$, the SNIPE estimator $\hat{\tau}_n^{\text{SNIPE}}$ is unbiased under D , and its worst-case mean squared error is at most*

$$B^2 \frac{dA_d}{n}.$$

- **(Local energy and binomial scale.)** For every unit i , the local SNIPE energy on either model class is bounded by A_d . Moreover, with $k_\star = k_\star(d, \beta, p)$ as in [definition 1](#),

$$c_{\beta,p} d \binom{d}{k_\star} \leq dA_d \leq C_{\beta,p} d \binom{d}{k_\star}.$$

- **(Exact block risk.)** If $d \geq 1$ and $d \mid n$, let $m = n/d$ and let the graph be the disjoint union of m complete directed d -blocks, including loops. Then canonical SNIPE has exact worst-case risk

$$\frac{B^2 A_d}{m}$$

on the fixed block graph for both model classes and globally over both model classes. Equivalently,

$$\frac{B^2 A_d}{m} = B^2 \frac{dA_d}{n}.$$

- **(Least-favourable block family.)** If $B > 0$ and $d \geq 1$, set $m = \lfloor n/d \rfloor$, $\rho = md/n$, and let δ be the perturbation amplitude. For every vector $U = (U_1, \dots, U_m)$ with $|U_b| \leq B/2$ for all b , both signed block schedules belong to $\mathcal{M}_{n,d,\beta}(B)$ on the complete-block graph, and their targets are

$$\rho\delta \quad \text{and} \quad -\rho\delta.$$

The squared Hellinger distance between the two corresponding block priors is bounded by

$$\frac{4\pi^2 m \delta^2}{B^2 A_d},$$

and

$$\frac{1}{2} \leq \rho \leq 1, \quad \frac{A_d}{m} = \frac{dA_d/n}{\rho}.$$

[Theorem 3](#) expresses the bounded-outcome comparison through the energy A_d , and [theorem 1](#) then converts that energy to the binomial scale indexed by $k_\star(d, \beta, p)$. The bounded-outcome lower frontier is inherited from the coefficient-mass lower frontier because the coefficient-mass class is contained carrier by carrier in the bounded-outcome class, giving $R_{n,\ell_1}^\star \leq R_{n,\infty}^\star$. The strict containment statement records how the two envelopes relate on the same graph carriers. The clipped estimator $\hat{\tau}_{n,\infty}^{\text{up}}$ attains the bounded-outcome rate after projection to the natural total-effect range, while the unprojected estimator remains unbiased and has worst-case mean squared error $B^2 d A_d / n$. On complete blocks with $d \mid n$, the theorem also gives the exact canonical SNIPE risk $B^2 A_d / m$, matching the local linear value in [theorem 2](#).

The final result specializes the energy calculation to fair-coin assignment. When $p = 1/2$, even-order Bernoulli contrasts vanish and odd orders determine the complete-block score energy. For first-order interactions ($\beta = 1$), this gives the transparent rate $B^2 \min\{1, d^2/n\}$, a useful benchmark for comparing low-order network interference estimators with the overlap-aware constructions in [Cortez-Rodriguez et al. \[2023\]](#) and subsequent degree-sensitive analyses [[Harshaw et al., 2022](#), [Eichhorn et al., 2024](#), [Wang and Li, 2026](#)].

Theorem 4. `[thm:fair-coin-energy-frontier]` (Fair-Coin Energy Frontier) *Let $p = 1/2$. The following statements hold.*

- **(Fair-coin contrasts and exposed order.)** For every $n, d, \beta \in \mathbb{N}$ and $B \in \mathbb{R}$ such that $n \geq 1$, $1 \leq d \leq n$, $\beta \geq 1$, and $B \geq 0$, the contrast coefficients and energy in [definition 1](#) satisfy, for every $r \geq 1$,

$$\Delta_r(1/2) = \begin{cases} 2^{1-r}, & r \text{ odd}, \\ 0, & r \text{ even}, \end{cases}$$

and

$$A_d = 4 \sum_{\substack{1 \leq r \leq \bar{\beta}_d \\ r \text{ odd}}} \binom{d}{r}.$$

Moreover, $k_*(d, \beta, 1/2)$ is the largest odd integer no greater than $\bar{\beta}_d$, with value 0 when no such odd integer exists.

- **(Degree-one minimax frontier.)** There exist constants $c_{\text{lower}}, c_{\text{upper}} \in \mathbb{R}$ with

$$0 < c_{\text{lower}} \leq c_{\text{upper}}$$

such that, for every $n, d \in \mathbb{N}$ and $B \in \mathbb{R}$ satisfying $n \geq 1$, $1 \leq d \leq n$, and $B \geq 0$, the degree-one energy is

$$A_d = 4d,$$

and the minimax risks in [definition 8](#) obey

$$c_{\text{lower}} B^2 \min\left\{1, \frac{d^2}{n}\right\} \leq R_n^*(d, 1, 1/2, B) = R_{n, \ell_1}^*(d, 1, 1/2, B) \leq R_{n, \infty}^*(d, 1, 1/2, B) \leq c_{\text{upper}} B^2 \min\left\{1, \frac{d^2}{n}\right\}.$$

If $d \mid n$, the unprojected SNIPE estimator satisfies the exact worst-case identities

$$\sup_{\mathcal{M}_{n, d, 1}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{4B^2 d^2}{n}$$

and

$$\sup_{\mathcal{M}_{n, d, 1}^\infty(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{4B^2 d^2}{n}.$$

[Theorem 4](#) makes the role of parity explicit. At $p = 1/2$, the exposed order is the largest odd interaction order up to $\bar{\beta}_d$, and the block energy is four times the corresponding sum of odd binomial coefficients. In the first-order interaction case ($\beta = 1$), $A_d = 4d$, so the degree factor in [theorem 3](#) becomes $dA_d = 4d^2$. The resulting minimax rate $B^2 \min\{1, d^2/n\}$ shows how first-order interference converts the graph degree into a quadratic design cost under fair-coin Bernoulli assignment.

4 Discussion and Extensions

The results above identify the design-based cost of total-effect estimation through a small set of finite-population quantities. The common feature in [theorems 1, 3](#) and [4](#) is the product of an overlap factor and a block energy. Under the bounded-degree condition, each treatment coordinate can enter at most d unit-level scores, and the overlap count in [theorem 1](#) turns this graph fact into the variance scale dA_d/n . The energy A_d then records how much Bernoulli

design information is required to recover the all-treated-versus-all-control contrast from low-order centered monomials.

This interpretation connects the rate to the exposure and overlap structure emphasized in network-experiment analyses [Leung, 2022, Sävje et al., 2021, Hu et al., 2022, Gao and Ding, 2025]. In the present model, the known graph fixes the neighborhoods and the low-order polynomial restriction fixes which assignment interactions are relevant. The exposed order $k_*(d, \beta, p)$ summarizes the maximum positive contrast order visible under Bernoulli assignment when such an order exists, with $k_* = 0$ under the degree-zero convention, and [theorem 1](#) converts that order into the binomial scale $d \binom{d}{k_*(d, \beta, p)} / n$. For fixed β and p , the rate therefore separates the sample-size contribution $1/n$, the graph-overlap contribution d , and the interaction-order contribution through the number of exposed subsets inside a degree- d neighborhood.

The closest estimator-level antecedent, Cortez-Rodriguez et al. [2023], develops SNIPE for low-order neighborhood interference under known-graph Bernoulli assignment and analyzes the resulting design variance through exposure overlap. The minimax frontier here turns that variance scale into a matched decision-theoretic characterization: the complete-block lower bound, the clipped SNIPE upper bound, and the binomial degree formula $dA_d/n \asymp_{\beta, p} d \binom{d}{k_*} / n$ all use the same block energy. The bounded-outcome result and the local-linear complete-block theorem provide the corresponding bounded-envelope and graph-local benchmarks.

The complete-block benchmark clarifies why the same quantity appears in both upper and lower bounds. In the least-favourable construction of [theorem 1](#), complete directed blocks align all within-block neighborhoods, so the block representer in [definition 9](#) generates the hardest local contrast at energy A_d . On this graph, [theorem 2](#) gives the exact local linear value $B^2 A_d / m_t$, equivalently $B^2 d_t A_d / n_t$. The global SNIPE upper bound attains the same order on arbitrary graphs because the score uses the same blockwise Riesz structure locally, while the bounded out-degree condition controls how often those local scores can share assignment coordinates.

The local linear characterization also separates risk optimality from a unique choice of weights. The canonical complete-block score is sufficient for attaining the benchmark, and [theorem 2](#) gives the exact blockwise quadratic criterion whose average excess characterizes asymptotic minimaxity inside the local linear class. Thus the complete-block calculation is more than a lower-bound device: it identifies the precise variance target that any asymptotically optimal local linear rule must match, while allowing multiple weight sequences to attain that target. This perspective complements recent work on low-order and degree-sensitive interference estimators [Wang and Li, 2026, Eichhorn et al., 2024].

The bounded-outcome extension in [theorem 3](#) shows that the same degree dependence is calibrated by A_d under a uniform potential-outcome envelope. The projection interval changes from the coefficient-mass scale to the natural total-effect range, but the unprojected SNIPE variance remains governed by $B^2 d A_d / n$, and clipping converts that variance calculation into the bounded risk comparison. The fair-coin specialization in [theorem 4](#) makes this calibration explicit: for $p = 1/2$, odd interaction orders determine A_d , and first-order interference yields the rate $B^2 \min\{1, d^2/n\}$.

Limitations and future work. The analysis is developed for known bounded-degree interference graphs, independent Bernoulli assignment with common treatment probability, and noiseless finite-population potential outcomes represented by low-order polynomial schedules. Natural extensions include alternative randomization schemes, covariate adjustment, graph uncertainty, clustered assignment, and sampling or outcome noise. These directions would connect

the minimax calculation here with broader design and exposure-mapping frameworks for network experiments [Leung, 2022, Sävje et al., 2021, Hu et al., 2022, Gao and Ding, 2025].

Appendices

A Proofs and Auxiliary Lemmas

This appendix collects the auxiliary finite-design facts that support the main risk comparisons. The first statement is the complete-block Riesz calculation used in the least-favourable construction and in the local linear benchmark. The second statement is the graph-counting identity that converts bounded in- and out-degree into the overlap scale appearing in [theorems 1](#) and [3](#). Together, these two ingredients isolate the two sources of the rate: block energy under Bernoulli assignment and neighborhood overlap in the graph.

Lemma 1. [`lem:block-energy-representer`] (Block Energy Representer) *Fix an integer $\beta \geq 1$ and $p \in (0, 1)$. With $k_\star(d, \beta, p)$ as in [definition 1](#), there exist constants $c_1, c_2, H \in \mathbb{R}$, depending on (β, p) , such that $0 < c_1 \leq c_2$ and $0 < H$, and the following statements hold for every integer $d \geq 1$:*

- **(Energy scale.)**

$$c_1 \binom{d}{k_\star(d, \beta, p)} \leq A_d \leq c_2 \binom{d}{k_\star(d, \beta, p)}.$$

- **(Representer identities.)** *For every finite design D on $\{0, 1\}^d$ satisfying the common- p product Bernoulli assignment condition in [assumption 1](#), and every $f \in \mathcal{P}_d$,*

$$\mathbb{E}_D[g_d(Z)f(Z)] = L_d(f).$$

For the same D ,

$$L_d(h_d) = 1, \quad \mathbb{E}_D[h_d(Z)^2] = A_d^{-1}.$$

- **(Perturbation program.)** *For every finite design D on $\{0, 1\}^d$ satisfying the common- p product Bernoulli assignment condition in [assumption 1](#), the representer h_d is feasible for the perturbation program in [definition 15](#). For every feasible perturbation h ,*

$$A_d^{-1} \leq \mathbb{E}_D[h(Z)^2],$$

with equality if and only if $h = h_d$.

- **(Weight program.)** *For every finite design D on $\{0, 1\}^d$ satisfying the common- p product Bernoulli assignment condition in [assumption 1](#), and every weight w feasible for the weight program in [definition 16](#) under D ,*

$$A_d \leq \mathbb{E}_D[w(Z)^2],$$

with equality if and only if $w = g_d$.

- **(Program values.)**

$$\text{PerturbProg}_{d, \beta, p} = A_d^{-1}, \quad \text{WeightProg}_{d, \beta, p} = A_d.$$

- (*Coefficient envelope.*)

$$\sum_{T \subseteq \{1, \dots, d\}} |h_{d,T}| \leq H.$$

Proof of lemma 1. Fix $\beta \geq 1$ and $p \in (0, 1)$. For every integer $r \geq 1$ write

$$e_r(p) := \frac{\Delta_r(p)^2}{[p(1-p)]^r}, \quad m_r(p) := \frac{|\Delta_r(p)|}{[p(1-p)]^r} (1+p)^r,$$

and define the three constants of the statement by

$$c_1 := \min\{e_r(p) : 1 \leq r \leq \beta, \Delta_r(p) \neq 0\}, \quad c_2 := 2^\beta \sum_{r=1}^{\beta} e_r(p), \quad H := c_1 + \frac{2^\beta \sum_{r=1}^{\beta} m_r(p)}{c_1}.$$

Since $\Delta_1(p) = (1-p) - (-p) = 1 \neq 0$, the index set defining c_1 contains $r = 1$ and is therefore a nonempty finite set; on it $\Delta_r(p)^2 > 0$ and $[p(1-p)]^r > 0$, so c_1 is a minimum of finitely many strictly positive numbers and $c_1 > 0$. Each $m_r(p) \geq 0$ for the same reason, so $H \geq c_1 > 0$.

Step 1 (exposed orders). Fix $d \geq 1$ and write $k_\star := k_\star(d, \beta, p)$. Since $\beta \geq 1$ and $d \geq 1$ we have $\bar{\beta}_d = \min\{\beta, d\} \geq 1$, and $\Delta_1(p) = 1 \neq 0$ shows that the set $\{r \in \{1, \dots, \bar{\beta}_d\} : \Delta_r(p) \neq 0\}$ of definition 1 is nonempty. Hence $k_\star$ is the maximum of that set, so

$$1 \leq k_\star \leq \bar{\beta}_d, \quad \Delta_{k_\star}(p) \neq 0, \quad \text{and} \quad r \leq k_\star \text{ for every } r \in \{1, \dots, \bar{\beta}_d\} \text{ with } \Delta_r(p) \neq 0.$$

In particular $k_\star \leq \beta$ and $k_\star \leq d$.

Step 2 (binomial domination at fixed order). For all integers $r \leq k \leq d$ with $k \leq \beta$,

$$\binom{d}{r} \leq 2^\beta \binom{d}{k}.$$

Indeed, counting the pairs $R \subseteq K \subseteq \{1, \dots, d\}$ with $|R| = r$ and $|K| = k$ in two ways gives $\binom{d}{r} \binom{d-r}{k-r} = \binom{d}{k} \binom{k}{r}$; the left factor $\binom{d-r}{k-r} \geq 1$ and $\binom{k}{r} \leq 2^k \leq 2^\beta$.

Step 3 (energy scale). By definition 9,

$$A_d = \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} e_r(p),$$

a sum of nonnegative terms. Discarding all summands except $r = k_\star$, which is legitimate by Step 1,

$$A_d \geq \binom{d}{k_\star} e_{k_\star}(p) \geq c_1 \binom{d}{k_\star},$$

the second inequality because $1 \leq k_\star \leq \beta$ and $\Delta_{k_\star}(p) \neq 0$, so $e_{k_\star}(p)$ is one of the numbers over which c_1 is the minimum. For the upper bound fix $r \in \{1, \dots, \bar{\beta}_d\}$. If $\Delta_r(p) = 0$ then $e_r(p) = 0$ and both sides of the next display vanish; if $\Delta_r(p) \neq 0$ then $r \leq k_\star \leq \min\{\beta, d\}$ by Step 1 and Step 2 applies. Either way

$$\binom{d}{r} e_r(p) \leq 2^\beta \binom{d}{k_\star} e_{k_\star}(p).$$

Summing over $r \in \{1, \dots, \bar{\beta}_d\}$ and then enlarging the index range from $\{1, \dots, \bar{\beta}_d\}$ to $\{1, \dots, \beta\}$, which only adds nonnegative terms since $\bar{\beta}_d \leq \beta$,

$$A_d \leq 2^\beta \binom{d}{k_\star} \sum_{r=1}^{\beta} e_r(p) = c_2 \binom{d}{k_\star}.$$

This is the energy-scale claim.

Step 4 ($c_1 \leq c_2$). Apply Step 3 at $d = 1$ and set $K := \binom{1}{k_\star(1, \beta, p)}$. Step 1 gives $k_\star(1, \beta, p) \leq \bar{\beta}_1 \leq 1$, so $K \geq 1 > 0$, and

$$c_1 K \leq A_1 \leq c_2 K.$$

Dividing by $K > 0$ gives $c_1 \leq c_2$.

Step 5 (design moments). Fix $d \geq 1$ and let D be a finite design on $\{0, 1\}^d$ satisfying the common- p product Bernoulli condition of [assumption 1](#). Then D assigns to each $z \in \{0, 1\}^d$ the mass $\prod_{j=1}^d p^{z_j} (1-p)^{1-z_j} > 0$, so for every $f : \{0, 1\}^d \rightarrow \mathbb{R}$,

$$\mathbb{E}_D[f(Z)^2] = 0 \quad \text{if and only if} \quad f(z) = 0 \quad \text{for every } z \in \{0, 1\}^d.$$

Moreover, factorizing the expectation over the independent coordinates, for every nonempty $S \subseteq \{1, \dots, d\}$ and every $T \subseteq \{1, \dots, d\}$,

$$\begin{aligned} \mathbb{E}_D \left[\prod_{j \in S} (Z_j - p) \prod_{j \in T} Z_j \right] &= \begin{cases} [p(1-p)]^{|S|} p^{|T|-|S|}, & S \subseteq T, \\ 0, & \text{otherwise,} \end{cases} \\ \mathbb{E}_D \left[\prod_{j \in S} (Z_j - p) \prod_{j \in T} (Z_j - p) \right] &= \begin{cases} [p(1-p)]^{|S|}, & S = T, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

In the first identity a coordinate in $S \cap T$ contributes $\mathbb{E}[(Z_j - p)Z_j] = p(1-p)$, a coordinate of T outside S contributes $\mathbb{E}[Z_j] = p$, a coordinate of S outside T contributes $\mathbb{E}[Z_j - p] = 0$, and all remaining coordinates contribute 1; in the second identity a coordinate in $S \cap T$ contributes $\mathbb{E}[(Z_j - p)^2] = p(1-p)$ and a coordinate of the symmetric difference contributes 0.

Step 6 (the score represents L_d). For every integer $t \geq 1$,

$$\sum_{r=1}^t \binom{t}{r} \Delta_r(p) p^{t-r} = 1,$$

because $\sum_{r=0}^t \binom{t}{r} (1-p)^r p^{t-r} = [(1-p) + p]^t = 1$ and $\sum_{r=0}^t \binom{t}{r} (-p)^r p^{t-r} = (-p + p)^t = 0$, while the two $r = 0$ terms cancel since $\Delta_0(p) = 0$. Let $T \subseteq \{1, \dots, d\}$ be nonempty with $|T| \leq \bar{\beta}_d$. Inserting the definition of g_d from [definition 9](#) into the first moment identity of Step 5, and using that exactly $\binom{|T|}{r}$ of the sets S with $|S| = r$ satisfy $S \subseteq T$,

$$\mathbb{E}_D \left[g_d(Z) \prod_{j \in T} Z_j \right] = \sum_{r=1}^{\bar{\beta}_d} \frac{\Delta_r(p)}{[p(1-p)]^r} \binom{|T|}{r} [p(1-p)]^r p^{|T|-r} = \sum_{r=1}^{\bar{\beta}_d} \binom{|T|}{r} \Delta_r(p) p^{|T|-r} = 1,$$

where the last equality is the previous display with $t = |T| \geq 1$: the summands with $|T| < r \leq \bar{\beta}_d$ vanish because $\binom{|T|}{r} = 0$, so the sum runs effectively over $1 \leq r \leq |T|$. Taking $T = \emptyset$ in the same moment identity, where no nonempty S satisfies $S \subseteq T$,

$$\mathbb{E}_D[g_d(Z)] = 0.$$

Now $L_d(z \mapsto \prod_{j \in T} z_j) = 1$ for nonempty T and $L_d(1) = 0$, so the two displays state precisely that $\mathbb{E}_D[g_d(Z)f(Z)] = L_d(f)$ whenever f is a raw monomial of degree at most $\bar{\beta}_d$. Both sides are linear in f , and by [definition 14](#) those monomials span $\mathcal{P}_d$; hence for every $f \in \mathcal{P}_d$,

$$\mathbb{E}_D[g_d(Z)f(Z)] = L_d(f).$$

Step 7 (energy identities and positivity). Evaluating g_d at the all-treated and at the all-control assignment, each of the $\binom{d}{r}$ sets S with $|S| = r$ contributes $(1-p)^r$ and $(-p)^r$ respectively, so

$$L_d(g_d) = \sum_{r=1}^{\bar{\beta}_d} \frac{\Delta_r(p)}{[p(1-p)]^r} \binom{d}{r} [(1-p)^r - (-p)^r] = \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} e_r(p) = A_d.$$

By the second moment identity of Step 5 the centered monomials appearing in g_d are pairwise orthogonal and the one indexed by S has second moment $[p(1-p)]^{|S|}$, so

$$\mathbb{E}_D[g_d(Z)^2] = \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^{2r}} [p(1-p)]^r = A_d.$$

Keeping only the $r = 1$ summand of A_d and using $\Delta_1(p) = 1$,

$$A_d \geq \binom{d}{1} \frac{\Delta_1(p)^2}{p(1-p)} = \frac{d}{p(1-p)} > 0.$$

Since $h_d = g_d/A_d$ with $A_d > 0$, dividing the two preceding identities by A_d and by A_d^2 gives

$$L_d(h_d) = 1, \quad \mathbb{E}_D[h_d(Z)^2] = A_d^{-1}.$$

Step 8 (the score lies in $\mathcal{P}_d$). Expanding the product coordinate by coordinate, for every $S \subseteq \{1, \dots, d\}$ and every z ,

$$\prod_{j \in S} (z_j - p) = \sum_{T \subseteq S} (-p)^{|S|-|T|} \prod_{j \in T} z_j.$$

Every raw monomial occurring on the right has degree $|T| \leq |S| \leq \bar{\beta}_d$ when $|S| \leq \bar{\beta}_d$, hence lies in $\mathcal{P}_d$. As g_d is a finite linear combination of centered monomials with $|S| \leq \bar{\beta}_d$, it follows that $g_d \in \mathcal{P}_d$ and therefore $h_d = g_d/A_d \in \mathcal{P}_d$. Together with $L_d(h_d) = 1$ from Step 7, h_d is feasible for the program of [definition 15](#).

Step 9 (perturbation optimality). Let $h \in \mathcal{P}_d$ satisfy $L_d(h) = 1$. Step 6 applied to $f = h$ gives $\mathbb{E}_D[g_d(Z)h(Z)] = 1$, and Step 7 gives $\mathbb{E}_D[g_d(Z)^2] = A_d$. Expanding the square,

$$\mathbb{E}_D[(A_d h(Z) - g_d(Z))^2] = A_d^2 \mathbb{E}_D[h(Z)^2] - 2A_d \mathbb{E}_D[g_d(Z)h(Z)] + \mathbb{E}_D[g_d(Z)^2] = A_d^2 \mathbb{E}_D[h(Z)^2] - A_d.$$

The left-hand side is nonnegative and $A_d > 0$, so $A_d^2 \mathbb{E}_D[h(Z)^2] \geq A_d$, that is,

$$A_d^{-1} \leq \mathbb{E}_D[h(Z)^2].$$

Equality holds if and only if the displayed square has expectation zero, which by the faithfulness statement of Step 5 happens if and only if $A_d h(z) = g_d(z)$ for every z , that is, if and only if $h = h_d$. Conversely $h = h_d$ attains the value A_d^{-1} by Step 7.

Step 10 (weight optimality). Let w be feasible for [definition 16](#) under D , that is, $\mathbb{E}_D[w(Z)] = 0$ and $\mathbb{E}_D[w(Z) \prod_{j \in S} Z_j] = 1$ for every nonempty $S \subseteq \{1, \dots, d\}$ with $|S| \leq \bar{\beta}_d$. Since $L_d(1) = 0$ and $L_d(z \mapsto \prod_{j \in S} z_j) = 1$ for nonempty S , these constraints say that $\mathbb{E}_D[w(Z)f(Z)] = L_d(f)$ on every raw monomial f of degree at most $\bar{\beta}_d$; both sides are linear in f , and those monomials span $\mathcal{P}_d$, so for every $f \in \mathcal{P}_d$,

$$\mathbb{E}_D[w(Z)f(Z)] = L_d(f).$$

Taking $f = g_d$, which lies in $\mathcal{P}_d$ by Step 8, and using $L_d(g_d) = A_d$ and $\mathbb{E}_D[g_d(Z)^2] = A_d$ from Step 7,

$$\mathbb{E}_D[(w(Z) - g_d(Z))^2] = \mathbb{E}_D[w(Z)^2] - 2\mathbb{E}_D[w(Z)g_d(Z)] + \mathbb{E}_D[g_d(Z)^2] = \mathbb{E}_D[w(Z)^2] - A_d.$$

Nonnegativity of the left-hand side gives

$$A_d \leq \mathbb{E}_D[w(Z)^2],$$

and equality forces the displayed square to have expectation zero, hence $w = g_d$ by the faithfulness statement of Step 5. Conversely, the two identities of Step 6 are exactly the feasibility constraints for $w = g_d$, and $\mathbb{E}_D[g_d(Z)^2] = A_d$, so g_d is feasible and attains the value A_d .

Step 11 (program values). Both programs are infima of sets of expectations of squares, hence of sets of nonnegative reals, and both sets are nonempty: h_d is feasible for the perturbation program with value A_d^{-1} by Steps 7 and 8, and g_d is feasible for the weight program with value A_d by Step 10. Therefore

$$\text{PerturbProg}_{d,\beta,p} \leq A_d^{-1}, \quad \text{WeightProg}_{d,\beta,p} \leq A_d,$$

while Steps 9 and 10 bound every value in the two feasible sets below by A_d^{-1} and by A_d respectively, giving the reverse inequalities. Hence

$$\text{PerturbProg}_{d,\beta,p} = A_d^{-1}, \quad \text{WeightProg}_{d,\beta,p} = A_d.$$

Step 12 (coefficient envelope). Expanding each centered monomial of g_d by the identity of Step 8, collecting the coefficient of $\prod_{j \in T} z_j$, and dividing by A_d , the raw coefficients of h_d in [definition 9](#) are

$$h_{d,T} = A_d^{-1} \sum_{r=1}^{\bar{\beta}_d} \sum_{\substack{S \subseteq \{1,\dots,d\}, \\ T \subseteq S, |S|=r}} \frac{\Delta_r(p)}{[p(1-p)]^r} (-p)^{r-|T|}.$$

Applying the triangle inequality inside each $h_{d,T}$ and then exchanging the order of summation, so that each S with $|S| = r$ is paired with the subsets $T \subseteq S$,

$$\sum_{T \subseteq \{1,\dots,d\}} |h_{d,T}| \leq A_d^{-1} \sum_{r=1}^{\bar{\beta}_d} \sum_{\substack{S \subseteq \{1,\dots,d\} \\ |S|=r}} \sum_{T \subseteq S} \frac{|\Delta_r(p)|}{[p(1-p)]^r} p^{r-|T|} = A_d^{-1} \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} m_r(p),$$

using $\sum_{T \subseteq S} p^{|S|-|T|} = (1+p)^{|S|} = (1+p)^r$ and the fact that there are $\binom{d}{r}$ sets S with $|S| = r$. Exactly as in Step 3 — orders with $\Delta_r(p) = 0$ contribute zero, orders with $\Delta_r(p) \neq 0$ satisfy

$r \leq k_\star$ and hence $\binom{d}{r} \leq 2^\beta \binom{d}{k_\star}$ by Step 2, and the index range may then be enlarged to $\{1, \dots, \beta\}$

$$\sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} m_r(p) \leq 2^\beta \binom{d}{k_\star} \sum_{r=1}^{\beta} m_r(p).$$

Combining this with the lower energy bound $c_1 \binom{d}{k_\star} \leq A_d$ of Step 3, with $A_d > 0$ and $\binom{d}{k_\star} > 0$,

$$\sum_{T \subseteq \{1, \dots, d\}} |h_{d,T}| \leq \frac{2^\beta \binom{d}{k_\star} \sum_{r=1}^{\beta} m_r(p)}{A_d} \leq \frac{2^\beta \sum_{r=1}^{\beta} m_r(p)}{c_1} \leq H,$$

the last step because H exceeds that quotient by $c_1 > 0$. $\square$

Lemma 1 supplies the finite-dimensional normalization used throughout the block arguments. The identity $\mathbb{E}_D[g_d(Z)f(Z)] = L_d(f)$ states that the complete-block score represents the all-one-versus-all-zero contrast on the low-order polynomial space from [definition 14](#). The perturbation and weight programs in [definitions 15](#) and [16](#) are therefore dual descriptions of the same Bernoulli information: the normalized perturbation has energy A_d^{-1} , while the unbiased block weight has energy A_d . The coefficient envelope gives a uniform way to embed the score-aligned perturbation inside the coefficient-mass model class in [definition 10](#). These calculations are the block analogue of inverse-probability and Horvitz–Thompson moment identities for randomized experiments [[Horvitz and Thompson, 1952](#), [Aronow and Samii, 2017](#)].

The next lemma records the overlap count used to pass from block-local variance identities to arbitrary bounded-degree graphs. It is purely combinatorial: a shared r -tuple of treatment coordinates contributes to the covariance of two unit-level scores only when that tuple lies in both neighborhoods, and the out-degree bound controls how often each tuple can be reused.

Lemma 2. [[lem:overlap-count](#)] (Overlap Count Bound) *Let G be a graph on V_n . Suppose that*

- (*Bounded degree.*) G satisfies [assumption 2](#) with degree bound d .
- (*Order.*) $r \in \mathbb{N}$ satisfies $1 \leq r$.
- (*Unit.*) $i \in V_n$.

Then

$$\sum_{l \in V_n} \binom{|N_i \cap N_l|}{r} = \sum_{\substack{S \subseteq N_i \\ |S|=r}} \#\{l \in V_n : S \subseteq N_l\} \leq d \binom{|N_i|}{r} \leq d \binom{d}{r}.$$

Proof of [lemma 2](#). Let

$$\mathcal{F} := \{S \subseteq N_i : |S| = r\}$$

be the family of r -element subsets of the neighborhood N_i , so that the middle expression in the statement is $\sum_{S \in \mathcal{F}} \#\{l \in V_n : S \subseteq N_l\}$.

Step 1: a pointwise counting identity. Fix $l \in V_n$. A finite set S satisfies $S \subseteq N_i \cap N_l$ if and only if $S \subseteq N_i$ and $S \subseteq N_l$, so the r -element subsets of $N_i \cap N_l$ are exactly the members of $\mathcal{F}$ that are contained in N_l :

$$\{S \subseteq N_i \cap N_l : |S| = r\} = \{S \in \mathcal{F} : S \subseteq N_l\}.$$

The number of r -element subsets of a finite set A is $\binom{|A|}{r}$. Taking cardinalities of the two sides and writing the cardinality of the right-hand side as a sum of indicators gives

$$\binom{|N_i \cap N_l|}{r} = \#\{S \in \mathcal{F} : S \subseteq N_l\} = \sum_{S \in \mathcal{F}} \mathbf{1}\{S \subseteq N_l\}.$$

Step 2: the double-count identity. Summing the display of Step 1 over $l \in V_n$ and exchanging the two finite sums,

$$\sum_{l \in V_n} \binom{|N_i \cap N_l|}{r} = \sum_{l \in V_n} \sum_{S \in \mathcal{F}} \mathbf{1}\{S \subseteq N_l\} = \sum_{S \in \mathcal{F}} \sum_{l \in V_n} \mathbf{1}\{S \subseteq N_l\} = \sum_{S \in \mathcal{F}} \#\{l \in V_n : S \subseteq N_l\},$$

the last equality because an indicator sum over $l \in V_n$ counts the l satisfying the condition. This is the equality asserted in the statement.

Step 3: each subset is reused at most d times. Fix $S \in \mathcal{F}$. Since $|S| = r$ and $1 \leq r$, the set S is nonempty; fix any $j \in S$. If $S \subseteq N_l$ then in particular $j \in N_l$, so

$$\{l \in V_n : S \subseteq N_l\} \subseteq \{l \in V_n : j \in N_l\}.$$

The out-degree half of [assumption 2](#), applied at the coordinate j , states that $\#\{l \in V_n : j \in N_l\} \leq d$. Combining this with monotonicity of cardinality under inclusion,

$$\#\{l \in V_n : S \subseteq N_l\} \leq \#\{l \in V_n : j \in N_l\} \leq d.$$

Step 4: the first inequality. The family $\mathcal{F}$ is precisely the family of r -element subsets of N_i , hence

$$|\mathcal{F}| = \binom{|N_i|}{r}.$$

Summing the bound of Step 3 over $S \in \mathcal{F}$ therefore gives

$$\sum_{S \in \mathcal{F}} \#\{l \in V_n : S \subseteq N_l\} \leq \sum_{S \in \mathcal{F}} d = d |\mathcal{F}| = d \binom{|N_i|}{r}.$$

Step 5: the second inequality. The in-degree half of [assumption 2](#), applied at the unit i , gives $|N_i| \leq d$. Since $m \mapsto \binom{m}{r}$ is nondecreasing in m , we get $\binom{|N_i|}{r} \leq \binom{d}{r}$, and multiplying this inequality by the nonnegative integer d yields

$$d \binom{|N_i|}{r} \leq d \binom{d}{r}.$$

Chaining the identity of Step 2 with the inequalities of Steps 4 and 5 proves the claim. $\square$

[Lemma 2](#) is the graph-theoretic step behind the SNIPE upper bound. After expanding products of centered Bernoulli monomials, only common neighborhood coordinates contribute nonzero covariance terms under [assumption 1](#). The equality rewrites the sum by first choosing the shared subset and then counting affected units; the final inequalities use the two degree bounds in [assumption 2](#). This converts local score energy into the global variance scale dA_d/n , matching the overlap-aware variance calculations used in network-interference estimators [[Cortez-Rodriguez et al., 2023](#), [Harshaw et al., 2022](#)].

The proof of [theorem 1](#) combines the two auxiliary lemmas. For the upper bound, the SNIPE score is unbiased for each low-order unit contrast by the same centered-monomial orthogonality encoded in [lemma 1](#); the covariance expansion is then summed with [lemma 2](#). Projection to $[-B, B]$ preserves the target scale of [definition 8](#) and turns the unprojected variance bound into the truncated risk comparison. For the lower bound, the complete-block family in [definition 10](#) uses h_d as the score-aligned perturbation, so the two signed priors separate the target by $\rho\delta$ while their affinity is controlled by the energy A_d^{-1} .

The proof of [theorem 2](#) evaluates the same block program inside the restricted local linear class. The moment equations defining [definition 18](#) reduce each block to the weight program in [definition 16](#), and [lemma 1](#) gives the minimum second moment A_{d_t} . The quadratic criterion $\Psi_{b,t}$ then records the exact blockwise worst-case contribution of admissible weights, yielding the finite-sample value and the asymptotic characterization stated in [theorem 2](#).

For the bounded-outcome comparison in [theorem 3](#), the coefficient-mass class embeds carrier by carrier into the uniformly bounded-outcome class, so the lower bound from [theorem 1](#) transfers through the risk ordering in [definition 22](#). The upper bound uses the same unbiased SNIPE variance calculation, with projection to $[-2B, 2B]$ as defined in [definition 23](#). On complete blocks, [lemma 1](#) gives the exact canonical variance $B^2 A_d/m$, aligning the global bounded-outcome comparison with the local linear benchmark.

Finally, [theorem 4](#) follows by evaluating the Bernoulli contrast coefficients at $p = 1/2$. Even-order contrasts cancel, odd-order contrasts remain, and the energy formula in [definition 9](#) reduces to the stated odd-binomial sum. In the first-order interaction case ($\beta = 1$), this gives $A_d = 4d$, so the general degree-dependent comparison in [theorem 3](#) specializes to the rate $B^2 \min\{1, d^2/n\}$.

B Verification Note

This appendix records the formal verification scope for the mathematical claims stated in the paper. The theorem statements and their proofs are machine-checked in Lean 4 for the finite Bernoulli design algebra underlying [assumption 1](#), the polynomial-interference constructions in [assumption 3](#) and [definition 5](#), the SNIPE unbiasedness and variance calculations associated with [definitions 6](#) and [7](#), and the minimax-risk definitions in [definitions 8](#) and [22](#). The same checked development covers the least-favourable complete-block family in [definition 10](#), the continuous-prior integrated-risk step, the affinity-based total-variation bound used for the block lower bound, and the block Riesz programs in [definitions 15](#) and [16](#).

The verification also covers the formal statements of [theorems 1](#) to [4](#) and the auxiliary statements collected in the proof appendix. The bibliographic inputs and econometric interpretation are supplied by the cited randomized-experiment, potential-outcomes, design-based, SNIPE, and low-order interference literatures [[Horvitz and Thompson, 1952](#), [Rubin, 1978](#), [Imbens and Rubin, 2015](#), [Aronow and Samii, 2017](#), [Cortez-Rodriguez et al., 2023](#), [Harshaw et al., 2022](#)].

C Proofs of the main results

Proof of [theorem 2](#). Throughout, when t is fixed we write $n = n_t$, $d = d_t$, $m = m_t$, $A = A_{d_t}$ and $\bar{\beta} = \bar{\beta}_{d_t} = \min\{\beta, d\}$. The population V_n is partitioned into the blocks $V_1, \dots, V_m$, each of size d ; $b(i)$ is the block containing i , $Z_b = (Z_j)_{j \in V_b}$, and $Z_S = \prod_{j \in S} Z_j$. Since G_t is the directed complete-block graph,

$$N_i^{G_t} = V_{b(i)} \quad \text{for every } i \in V_n, \quad n = md.$$

The clauses are proved in the order in which their ingredients are established: the exact worst-case formula, then the per-block lower bound, then the minimax constant, then the two asymptotic criteria, then the witness sequence.

We first record the three block-score identities that every step uses. The block score g_d of [definition 9](#) lies in $\mathcal{P}_d$, and [lemma 1](#) gives, under [assumption 1](#),

$$\begin{aligned}\mathbb{E}_Z[g_d(Z_b)] &= 0, & \mathbb{E}_Z[g_d(Z_b)Z_S] &= 1 \quad \text{for every } \emptyset \neq S \subseteq V_b \text{ with } |S| \leq \bar{\beta}, \\ \mathbb{E}_Z[g_d(Z_b)^2] &= L_d(g_d) = A > 0.\end{aligned}$$

The first two displays are the representer identity $\mathbb{E}_Z[g_d(Z)f(Z)] = L_d(f)$ at $f \equiv 1$ and at $f = Z_S$; the third combines that identity at $f = g_d$ with $\mathbb{E}_Z[h_d(Z)^2] = A^{-1}$ and $g_d = Ah_d$, and the strict positivity is the energy-scale bound $A \geq c_1 \binom{d}{k_\star(d, \beta, p)} > 0$.

1. **The exact worst-case risk formula.** Fix t and $w \in \mathcal{E}_t^{\text{loc}, \text{lin}}$. For $c \in \mathcal{M}_t(G_t)$ the potential outcomes are $Y_i(Z) = \sum_{S \subseteq V_{b(i)}, |S| \leq \bar{\beta}} c_{i,S} Z_S$, and the target is the all-treated versus all-control contrast

$$\tau_n = \frac{1}{n} \sum_{i \in V_n} \{Y_i(\mathbf{1}_n) - Y_i(\mathbf{0}_n)\} = \frac{1}{n} \sum_{i \in V_n} \sum_{\substack{S \subseteq V_{b(i)} \\ |S| \leq \bar{\beta}}} c_{i,S} \mathbf{1}(S \neq \emptyset).$$

Subtracting this from $\hat{\tau}_{w,t} = n^{-1} \sum_i w_{i,t}(Z_{b(i)})Y_i(Z)$ and grouping units by block,

$$\hat{\tau}_{w,t} - \tau_n = \frac{1}{n} \sum_{b=1}^m e_b(c; Z), \quad e_b(c; Z) := \sum_{i \in V_b} \sum_{\substack{S \subseteq V_b \\ |S| \leq \bar{\beta}}} c_{i,S} \{w_{i,t}(Z_b)Z_S - \mathbf{1}(S \neq \emptyset)\}.$$

Each $e_b(c; Z)$ is a function of Z_b alone, and the moment restrictions defining $\mathcal{E}_t^{\text{loc}, \text{lin}}$ give $\mathbb{E}_Z[e_b(c; Z)] = 0$ termwise. Distinct blocks use disjoint assignment coordinates, so under [assumption 1](#) the variables $e_1(c; Z), \dots, e_m(c; Z)$ are independent and the cross terms vanish:

$$\mathbb{E}_Z[(\hat{\tau}_{w,t} - \tau_n)^2] = \frac{1}{n^2} \sum_{b=1}^m \mathbb{E}_Z[e_b(c; Z)^2].$$

For a fixed block, $c \mapsto e_b(c; Z)$ is linear in the rows $(c_{i,S})_S$ and the only constraint $\mathcal{M}_t(G_t)$ places on a row is $\sum_S |c_{i,S}| \leq B$ with $S \subseteq V_b$ and $|S| \leq \bar{\beta}$. The criterion $c \mapsto \mathbb{E}_Z[e_b(c; Z)^2]$ is convex in each row, so its maximum over the row simplex of radius B is attained at a vertex $c_{i,S} = \varepsilon_i B \mathbf{1}(S = S_i)$ with $\varepsilon_i \in \{-1, 1\}$, for which $e_b(c; Z) = B \sum_{i \in V_b} \varepsilon_i \{w_{i,t}(Z_b)Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\}$. Hence

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[e_b(c; Z)^2] = B^2 \Psi_{b,t}(w).$$

The row constraint is unitwise, so the maximizing vertices may be chosen block by block and assembled into one schedule in $\mathcal{M}_t(G_t)$ that attains every block maximum simultaneously. Therefore

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w,t} - \tau_n)^2] = \frac{B^2}{n^2} \sum_{b=1}^m \Psi_{b,t}(w),$$

which is the asserted worst-case identity.

2. **The per-block lower bound.** Fix $t, w \in \mathcal{E}_t^{\text{loc}, \text{lin}}$ and a block V_b . Every $i \in V_b$ has $N_i^{G_t} = V_b$, so the single function $g_d(Z_b)$ is the common block score for all of them. Expand it in the raw monomials it is spanned by,

$$g_d(Z_b) = \sum_{\substack{T \subseteq V_b \\ |T| \leq \bar{\beta}}} a_T Z_T, \quad \sum_{\substack{T \subseteq V_b \\ |T| \leq \bar{\beta}}} a_T = g_d(\mathbf{1}_d), \quad a_\emptyset = g_d(\mathbf{0}_d).$$

Applying $\mathbb{E}_Z[w_{i,t}] = 0$ to the term $T = \emptyset$ and $\mathbb{E}_Z[w_{i,t}Z_T] = 1$ to every other term gives, for every $i \in V_b$,

$$\mathbb{E}_Z[w_{i,t}(Z_b)g_d(Z_b)] = \sum_{\substack{\emptyset \neq T \subseteq V_b \\ |T| \leq \bar{\beta}}} a_T = g_d(\mathbf{1}_d) - g_d(\mathbf{0}_d) = L_d(g_d) = A.$$

Summing over the d units of V_b and using $\mathbb{E}_Z[g_d(Z_b)^2] = A$, the expansion of a nonnegative square gives

$$0 \leq \mathbb{E}_Z \left[\left\{ \sum_{i \in V_b} w_{i,t}(Z_b) - dg_d(Z_b) \right\}^2 \right] = \mathbb{E}_Z \left[\left\{ \sum_{i \in V_b} w_{i,t}(Z_b) \right\}^2 \right] - 2d(dA) + d^2A,$$

that is,

$$d^2A \leq \mathbb{E}_Z \left[\left\{ \sum_{i \in V_b} w_{i,t}(Z_b) \right\}^2 \right].$$

The choice $\varepsilon_i = 1$ and $S_i = \emptyset$ for every $i \in V_b$ is admissible in $\Psi_{b,t}$, since $\emptyset \subseteq V_b$ and $|\emptyset| = 0 \leq \bar{\beta}$, and it produces exactly that expectation; as $\Psi_{b,t}$ is a maximum over admissible choices,

$$\mathbb{E}_Z \left[\left\{ \sum_{i \in V_b} w_{i,t}(Z_b) \right\}^2 \right] \leq \Psi_{b,t}(w), \quad \text{hence} \quad d_t^2 A_{d_t} \leq \Psi_{b,t}(w).$$

3. **The exact minimax constant.** Define the canonical block-local weights

$$w_{i,t}^0(Z) := g_d(Z_{b(i)}),$$

which are the SNIPE scores of [definition 6](#) on G_t because $N_i^{G_t} = V_{b(i)}$ has exactly d elements. They depend on Z only through $Z_{b(i)}$, and the three block-score identities recorded above are precisely the centering and unit raw-moment restrictions, so $w^0 \in \mathcal{E}_t^{\text{loc}, \text{lin}}$.

Fix a block V_b and an admissible choice $(\varepsilon_i, S_i)_{i \in V_b}$ in $\Psi_{b,t}$. All units of V_b carry the same weight $g_d(Z_b)$, so

$$\sum_{i \in V_b} \varepsilon_i \{w_{i,t}^0(Z_b)Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\} = g_d(Z_b)F(Z_b) - \Lambda,$$

$$F(Z_b) := \sum_{i \in V_b} \varepsilon_i Z_{S_i}, \quad \Lambda := \sum_{i \in V_b} \varepsilon_i \mathbf{1}(S_i \neq \emptyset).$$

The block-score identities give $\mathbb{E}_Z[g_d(Z_b)F(Z_b)] = \Lambda$, whence

$$\mathbb{E}_Z \left[(g_d(Z_b)F(Z_b) - \Lambda)^2 \right] = \mathbb{E}_Z \left[(g_d(Z_b)F(Z_b))^2 \right] - \Lambda^2 \leq \mathbb{E}_Z \left[(g_d(Z_b)F(Z_b))^2 \right].$$

Since $0 \leq Z_{S_i} \leq 1$, $|\varepsilon_i| = 1$ and $|V_b| = d$, we have $|F(Z_b)| \leq d$ pointwise, so

$$\mathbb{E}_Z \left[(g_d(Z_b)F(Z_b))^2 \right] \leq d^2 \mathbb{E}_Z[g_d(Z_b)^2] = d^2 A.$$

Maximizing over admissible choices and combining with the preceding item,

$$\Psi_{b,t}(w^0) \leq d^2 A, \quad \text{hence} \quad \Psi_{b,t}(w^0) = d^2 A.$$

By the worst-case identity and the per-block lower bound, every $w \in \mathcal{E}_t^{\text{loc,lin}}$ satisfies

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w,t} - \tau_n)^2] = \frac{B^2}{n^2} \sum_{b=1}^m \Psi_{b,t}(w) \geq \frac{B^2}{n^2} m d^2 A = \frac{B^2 A}{m},$$

the last equality because $n = md$; and the same computation with $\Psi_{b,t}(w^0) = d^2 A$ turns the inequality into an equality for w^0 . Taking the infimum over $\mathcal{E}_t^{\text{loc,lin}}$,

$$R_t^{\text{loc,lin}} = \frac{B^2 A_{d_t}}{m_t} = \frac{B^2 d_t A_{d_t}}{n_t},$$

where the second equality is again $n_t = m_t d_t$.

4. **The risk ratio equals one plus the normalized block excess.** For a sequence $w_t \in \mathcal{E}_t^{\text{loc,lin}}$ set

$$Y_t := \frac{1}{m_t d_t^2 A_{d_t}} \sum_{b=1}^{m_t} \{ \Psi_{b,t}(w_t) - d_t^2 A_{d_t} \}.$$

Dividing the worst-case identity by $B^2 A_{d_t}/m_t$ and using $n_t^2 = m_t^2 d_t^2$,

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} = \frac{m_t}{n_t^2 A_{d_t}} \sum_{b=1}^{m_t} \Psi_{b,t}(w_t) = \frac{1}{m_t d_t^2 A_{d_t}} \sum_{b=1}^{m_t} \Psi_{b,t}(w_t) = 1 + Y_t.$$

Adding, respectively subtracting, the constant sequence 1 therefore shows that the left-hand ratio converges to 1 if and only if $Y_t \rightarrow 0$, which is the asserted equivalence.

5. **Score closeness implies attainment.** Put $\delta_{i,t} := w_{i,t} - g_{d_t}$ on the block of i , and set

$$E_{b,t} := \sum_{i \in V_b} \mathbb{E}_Z[\delta_{i,t}(Z_b)^2], \quad E_t := \sum_{b=1}^{m_t} E_{b,t} = \sum_{i \in V_{n_t}} \mathbb{E}_Z[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2],$$

$$X_t := \frac{E_t}{n_t A_{d_t}} = \frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2],$$

so that the hypothesis of this clause is $X_t \rightarrow 0$. The per-block lower bound gives $Y_t \geq 0$ for every t .

Fix t , a block V_b and an admissible choice $(\varepsilon_i, S_i)_{i \in V_b}$. Splitting each weight into its canonical part and its deviation,

$$\sum_{i \in V_b} \varepsilon_i \{w_{i,t}(Z_b) Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\} = f(Z_b) + v(Z_b),$$

$$f(Z_b) := \sum_{i \in V_b} \varepsilon_i \{g_{d_t}(Z_b) Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\}, \quad v(Z_b) := \sum_{i \in V_b} \varepsilon_i \delta_{i,t}(Z_b) Z_{S_i}.$$

The canonical bound of the third item applies to f , and for v the Cauchy–Schwarz inequality over the d_t summands together with $0 \leq Z_{S_i} \leq 1$ and $|\varepsilon_i| = 1$ gives $v(Z_b)^2 \leq d_t \sum_{i \in V_b} \delta_{i,t}(Z_b)^2$ pointwise, so

$$\mathbb{E}_Z[f(Z_b)^2] \leq d_t^2 A_{d_t}, \quad \mathbb{E}_Z[v(Z_b)^2] \leq d_t E_{b,t}.$$

Bounding the cross term by Cauchy–Schwarz, $\mathbb{E}_Z[(f+v)^2] \leq \mathbb{E}_Z[f^2] + 2\sqrt{\mathbb{E}_Z[f^2]}\sqrt{\mathbb{E}_Z[v^2]} + \mathbb{E}_Z[v^2]$, and maximizing over admissible choices,

$$\Psi_{b,t}(w_t) - d_t^2 A_{d_t} \leq 2\sqrt{d_t^2 A_{d_t}} \sqrt{d_t E_{b,t}} + d_t E_{b,t}.$$

Summing over b and applying Cauchy–Schwarz once more, $\sum_{b=1}^{m_t} \sqrt{d_t E_{b,t}} \leq \sqrt{m_t} \sqrt{d_t E_t}$, so

$$\sum_{b=1}^{m_t} \{\Psi_{b,t}(w_t) - d_t^2 A_{d_t}\} \leq 2\sqrt{d_t^2 A_{d_t}} \sqrt{m_t} \sqrt{d_t E_t} + d_t E_t.$$

Dividing by $m_t d_t^2 A_{d_t}$ and using $n_t = m_t d_t$,

$$\frac{2\sqrt{d_t^2 A_{d_t}} \sqrt{m_t} \sqrt{d_t E_t} + d_t E_t}{m_t d_t^2 A_{d_t}} = 2\sqrt{\frac{E_t}{n_t A_{d_t}}} + \frac{E_t}{n_t A_{d_t}} = 2\sqrt{X_t} + X_t,$$

so that

$$0 \leq Y_t \leq 2\sqrt{X_t} + X_t.$$

If $X_t \rightarrow 0$ then the right-hand bound tends to 0 by continuity of the square root, so $Y_t \rightarrow 0$ by squeezing, and the identity of the preceding item turns this into the asserted convergence of the risk ratio to 1.

6. **An optimal sequence at normalized distance two.** Assume $d_t \rightarrow \infty$. Then $\beta < d_t$ for all sufficiently large t , and for such t we have $\bar{\beta}_{d_t} = \beta$. Fix, for each block V_b , a distinguished unit $i_b^* \in V_b$, write

$$\psi_b(Z_b) := \left\{ \frac{2d_t A_{d_t}}{[p(1-p)]^{d_t}} \right\}^{1/2} \prod_{j \in V_b} (Z_j - p),$$

and define

$$w_{i,t}(Z) := g_{d_t}(Z_{b(i)}) + \mathbf{1}(i = i_{b(i)}^*) \psi_{b(i)}(Z_{b(i)}).$$

Admissibility. Each ψ_b is a function of Z_b alone. For every $S \subseteq V_b$ with $|S| \leq \beta < d_t$, including $S = \emptyset$, at least one index $j \in V_b$ outside S occurs in the product, and under

[assumption 1](#) that coordinate is independent of Z_S and of the remaining factors and satisfies $\mathbb{E}_Z[Z_j - p] = 0$; hence

$$\mathbb{E}_Z \left[\left(\prod_{j \in V_b} (Z_j - p) \right) Z_S \right] = 0 \quad \text{for every } S \subseteq V_b \text{ with } |S| < d_t.$$

So the perturbation changes neither the centering nor the unit raw moments inherited from g_{d_t} , and $w_t \in \mathcal{E}_t^{\text{loc,lin}}$.

Perturbation energy. By independence and $\mathbb{E}_Z[(Z_j - p)^2] = p(1 - p)$,

$$\mathbb{E}_Z \left[\left(\prod_{j \in V_b} (Z_j - p) \right)^2 \right] = [p(1 - p)]^{d_t}, \quad \text{hence} \quad \mathbb{E}_Z[\psi_b(Z_b)^2] = 2d_t A_{d_t}.$$

Block bound and the risk ratio. Fix a block V_b and an admissible choice, and split the inner sum as in the preceding item. Because the perturbation is carried by the single unit i_b^* , the deviation part reduces to one summand, $v(Z_b) = \varepsilon_{i_b^*} \psi_b(Z_b) Z_{S_{i_b^*}}$, and $0 \leq Z_{S_{i_b^*}} \leq 1$ gives $\mathbb{E}_Z[v(Z_b)^2] \leq \mathbb{E}_Z[\psi_b(Z_b)^2] = 2d_t A_{d_t}$; the canonical part still obeys $\mathbb{E}_Z[f(Z_b)^2] \leq d_t^2 A_{d_t}$. The same Cauchy–Schwarz bound on the cross term therefore yields, for all sufficiently large t and every b ,

$$\Psi_{b,t}(w_t) \leq d_t^2 A_{d_t} + 2\sqrt{d_t^2 A_{d_t}} \sqrt{2d_t A_{d_t}} + 2d_t A_{d_t}.$$

Dividing the excess by $d_t^2 A_{d_t}$ gives $2\sqrt{2/d_t} + 2/d_t$ for each block, hence the same bound for Y_t , while $Y_t \geq 0$ by the per-block lower bound. With the identity of the fourth item,

$$1 \leq \frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t} / m_t} \leq 1 + 2\sqrt{\frac{2}{d_t}} + \frac{2}{d_t}$$

for all sufficiently large t . Since $d_t \rightarrow \infty$, squeezing proves the asserted convergence of the risk ratio to 1.

Normalized distance. For the same sufficiently large t , the deviation $w_{i,t} - g_{d_t}$ equals ψ_b at $i = i_b^*$ and vanishes at every other unit, so

$$\sum_{i \in V_{n_t}} \mathbb{E}_Z \left[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2 \right] = \sum_{b=1}^{m_t} \mathbb{E}_Z[\psi_b(Z_b)^2] = 2m_t d_t A_{d_t},$$

and dividing by $n_t A_{d_t} = m_t d_t A_{d_t}$ gives the value 2.

Invariance under block-preserving relabeling. Let π be a bijection of V_{n_t} with $b(\pi(i)) = b(i)$ for every i ; then π maps each block onto itself and therefore permutes the coordinates of Z_b among themselves. The relabeled distance moves the score term only: at the unit i it compares the weight $w_{i,t}$, read at the unit i and at the original assignment, with the canonical score of the relabeled unit $\pi(i)$, whose neighborhood is again $V_{b(i)}$, read at the relabeled assignment whose j th coordinate is $Z_{\pi^{-1}(j)}$. Since g_{d_t} is a symmetric function of the coordinates of its block, that relabeled score is the original one,

$$g_{d_t}((Z_{\pi^{-1}(j)})_{j \in V_{b(i)}}) = g_{d_t}(Z_{b(i)}) \quad \text{for every } i \in V_{n_t}.$$

Each summand of the normalized distance is therefore unchanged, and the previous display gives

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z \left[\left\{ w_{i,t}(Z_{b(i)}) - g_{d_t}((Z_{\pi^{-1}(j)})_{j \in V_{b(i)}}) \right\}^2 \right] = 2,$$

which is the asserted invariance. □

Proof of [theorem 4](#). 1. **Fair-coin contrasts.** Fix $n, d, \beta \in \mathbb{N}$ and $B \in \mathbb{R}$ with $n \geq 1$, $1 \leq d \leq n$, $\beta \geq 1$, and $B \geq 0$; the three identities of this and the next two steps involve only d and β . The contrast coefficient is $\Delta_r(p) = (1-p)^r - (-p)^r$, so at $p = 1/2$ every $r \geq 1$ satisfies

$$\Delta_r(1/2) = \left(\frac{1}{2}\right)^r - \left(-\frac{1}{2}\right)^r = 2^{-r} - (-1)^r 2^{-r} = 2^{-r} \{1 - (-1)^r\}.$$

Since $1 - (-1)^r$ equals 2 for odd r and 0 for even r ,

$$\Delta_r(1/2) = \begin{cases} 2^{1-r}, & r \text{ odd}, \\ 0, & r \text{ even}. \end{cases}$$

2. **Fair-coin energy.** By [definition 9](#) at $p = 1/2$, the design variance factor is $[p(1-p)]^r = (1/4)^r$, so

$$A_d = \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \frac{\Delta_r(1/2)^2}{\left[\frac{1}{2} \left(1 - \frac{1}{2}\right)\right]^r}.$$

Each summand is evaluated by the previous step: for every r with $1 \leq r \leq \bar{\beta}_d$,

$$\binom{d}{r} \frac{\Delta_r(1/2)^2}{\left[\frac{1}{2} \left(1 - \frac{1}{2}\right)\right]^r} = \begin{cases} 4 \binom{d}{r}, & r \text{ odd}, \\ 0, & r \text{ even}, \end{cases}$$

because for odd r the numerator is $\binom{d}{r} (2^{1-r})^2 = 4 \binom{d}{r} 4^{-r}$ while the denominator is $(1/4)^r = 4^{-r}$, and for even r the numerator vanishes. Summing the surviving terms,

$$A_d = 4 \sum_{\substack{1 \leq r \leq \bar{\beta}_d \\ r \text{ odd}}} \binom{d}{r}.$$

3. **Fair-coin exposed order.** The index set in [definition 1](#) satisfies

$$\{r \in \{1, \dots, \bar{\beta}_d\} : \Delta_r(1/2) \neq 0\} = \{r \in \{1, \dots, \bar{\beta}_d\} : r \text{ odd}\},$$

since $\Delta_r(1/2) = 2^{1-r} \neq 0$ for odd r and $\Delta_r(1/2) = 0$ for even r . Hence $k_*(d, \beta, 1/2)$ is the maximum of the odd integers in $\{1, \dots, \bar{\beta}_d\}$ when that set is nonempty, and equals 0 otherwise, which is the value assigned by the zero-degree convention in [definition 1](#).

4. **Choice of constants.** Apply [theorem 3](#) with interaction order $\beta = 1$, which satisfies $\beta \geq 1$, and treatment probability $p = 1/2$, which satisfies $0 < p < 1$. It supplies constants $c_{1,1/2}$ and $C_{1,1/2}$ with

$$0 < c_{1,1/2} \leq C_{1,1/2}.$$

Set

$$c_{\text{lower}} := c_{1,1/2}, \quad c_{\text{upper}} := 4C_{1,1/2}.$$

Then $c_{\text{lower}} > 0$, and since $0 < C_{1,1/2}$ gives $C_{1,1/2} \leq 4C_{1,1/2}$,

$$0 < c_{\text{lower}} \leq C_{1,1/2} \leq 4C_{1,1/2} = c_{\text{upper}}.$$

5. **Design.** Fix $n, d \in \mathbb{N}$ and $B \in \mathbb{R}$ with $n \geq 1$, $1 \leq d \leq n$, and $B \geq 0$, and take for P_Z the product Bernoulli law on V_n with common treatment probability $1/2$, that is,

$$\Pr(Z_j = 1) = \frac{1}{2} \quad \text{for every } j \in V_n,$$

with $\{Z_j : j \in V_n\}$ mutually independent. Since $0 < 1/2 < 1$, this design satisfies [assumption 1](#) at $p = 1/2$, so the conclusions of [theorem 3](#) are available at (n, d, B) with $\beta = 1$ and $p = 1/2$.

6. **Degree-one energy.** With $\beta = 1$ and $d \geq 1$ the exposed order of [definition 1](#) is

$$\bar{\beta}_d = \min\{1, d\} = 1,$$

so the odd integers in $\{1, \dots, \bar{\beta}_d\}$ are the single index $r = 1$. The energy identity of Step 2 therefore gives

$$A_d = 4 \binom{d}{1} = 4d.$$

7. **Reduction of the rate argument.** Put

$$x := \frac{d^2}{n}.$$

Since $n \geq 1$, we have $x \geq 0$, and by the previous step

$$\frac{dA_d}{n} = \frac{d(4d)}{n} = 4x.$$

Two elementary comparisons of truncated arguments are used below. First, $x \leq 4x$ because $x \geq 0$, and truncation at 1 is monotone, so

$$\min\{1, x\} \leq \min\{1, 4x\}.$$

Second,

$$\min\{1, 4x\} \leq 4 \min\{1, x\}.$$

Indeed, if $x \leq 1$ then $\min\{1, x\} = x$, and either $4x \leq 1$, in which case $\min\{1, 4x\} = 4x = 4 \min\{1, x\}$, or $4x > 1$, in which case $\min\{1, 4x\} = 1 \leq 4x = 4 \min\{1, x\}$; while if $x > 1$ then $\min\{1, x\} = 1$ and $\min\{1, 4x\} \leq 1 \leq 4 = 4 \min\{1, x\}$.

8. **Lower bound.** Because $c_{\text{lower}} > 0$ and $B \geq 0$,

$$0 \leq c_{\text{lower}} B^2,$$

so multiplying the first comparison of Step 7 by this factor preserves the inequality. The lower frontier bound of [theorem 3](#) at $\beta = 1$, $p = 1/2$ reads

$$c_{\text{lower}} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\} \leq R_{n,\ell_1}^*(d, 1, 1/2, B),$$

and the same result identifies

$$R_{n,\ell_1}^*(d, 1, 1/2, B) = R_n^*(d, 1, 1/2, B).$$

Chaining these with the two identities of Step 7,

$$c_{\text{lower}} B^2 \min \left\{ 1, \frac{d^2}{n} \right\} = c_{\text{lower}} B^2 \min \{1, x\} \leq c_{\text{lower}} B^2 \min \{1, 4x\} = c_{\text{lower}} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\} \leq R_n^*(d, 1, 1/2, B)$$

9. **Class comparison.** The carrier inclusion recorded in [theorem 3](#) supplies

$$R_{n,\ell_1}^*(d, 1, 1/2, B) \leq R_{n,\infty}^*(d, 1, 1/2, B).$$

10. **Upper bound.** The clipped bounded-outcome estimator $\hat{\tau}_{n,\infty}^{\text{up}}$ of [definition 23](#) is one admissible competitor in the bounded-outcome minimax problem, and [theorem 3](#) bounds its worst-case risk, giving

$$R_{n,\infty}^*(d, 1, 1/2, B) \leq \sup_{\mathcal{M}_{n,d,1}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2 \right] \leq C_{1,1/2} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

Since $C_{1,1/2} > 0$ and $B \geq 0$, the factor $C_{1,1/2} B^2$ is nonnegative, so the second comparison of Step 7 may be multiplied by it. With $dA_d/n = 4x$ this yields

$$C_{1,1/2} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\} = C_{1,1/2} B^2 \min \{1, 4x\} \leq 4C_{1,1/2} B^2 \min \{1, x\} = c_{\text{upper}} B^2 \min \left\{ 1, \frac{d^2}{n} \right\},$$

and therefore

$$R_{n,\infty}^*(d, 1, 1/2, B) \leq c_{\text{upper}} B^2 \min \left\{ 1, \frac{d^2}{n} \right\}.$$

Steps 8–10 together give the asserted chain

$$c_{\text{lower}} B^2 \min \left\{ 1, \frac{d^2}{n} \right\} \leq R_n^*(d, 1, 1/2, B) = R_{n,\ell_1}^*(d, 1, 1/2, B) \leq R_{n,\infty}^*(d, 1, 1/2, B) \leq c_{\text{upper}} B^2 \min \left\{ 1, \frac{d^2}{n} \right\}.$$

11. **Exact complete-block identities.** Assume in addition that $d \mid n$ and set $m := n/d$, the number of complete d -blocks. The exact-block-risk clause of [theorem 3](#) gives, for the unprojected SNIPE estimator of [definition 7](#),

$$\sup_{\mathcal{M}_{n,d,1}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{B^2 A_d}{m} = \sup_{\mathcal{M}_{n,d,1}^\infty(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right],$$

together with the identity

$$\frac{B^2 A_d}{m} = \frac{B^2 d A_d}{n}.$$

Substituting $A_d = 4d$ from Step 6,

$$\frac{B^2 d A_d}{n} = \frac{B^2 d (4d)}{n} = \frac{4B^2 d^2}{n},$$

which is the asserted common value of both worst-case risks. $\square$

Proof of [theorem 1](#). 1. **The two constants.** Apply [theorem 3](#) at the fixed $\beta \geq 1$ and $p \in (0, 1)$, and write a and b for the pair of constants it produces. That result calls them $c_{\beta,p}$ and $C_{\beta,p}$; they are renamed here because the present statement reserves those two names for the constants constructed in this proof. They satisfy

$$0 < a \leq b.$$

Define

$$c_{\beta,p} := \min\{1, a^2\}, \quad C_{\beta,p} := \max\{1, b^2\}.$$

Since $a > 0$ gives $a^2 > 0$, the number $\min\{1, a^2\}$ is positive, and $\min\{1, a^2\} \leq 1 \leq \max\{1, b^2\}$. Hence

$$0 < c_{\beta,p} \leq C_{\beta,p}.$$

2. **The clauses that are used.** Fix $n, d \in \mathbb{N}$ and $B \in \mathbb{R}$ with $n \geq 1$, $d \leq n$, and $B \geq 0$, write $k_\star = k_\star(d, \beta, p)$, and set

$$x := \frac{d \binom{d}{k_\star}}{n}, \quad y := \frac{d A_d}{n},$$

so that $x \geq 0$. The assignment law is the common-probability product Bernoulli design with probability p , so it satisfies [assumption 1](#), and [theorem 3](#) therefore applies at this population size, this design, this degree bound, and this envelope. Of its conclusions the argument below uses exactly five. First, the lower frontier on the energy scale, together with the identity $R_{n,\ell_1}^\star = R_n^\star$ of [definition 22](#),

$$a B^2 \min\{1, y\} \leq R_{n,\ell_1}^\star(d, \beta, p, B) = R_n^\star(d, \beta, p, B).$$

Second, the projected-SNIPE upper bound on the energy scale,

$$\sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] \leq b B^2 \min\{1, y\}.$$

Third, the unprojected-SNIPE bound, which carries no constant,

$$\sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] \leq B^2 y.$$

Fourth, the local-energy comparison: for every $(G, c) \in \mathcal{M}_{n,d,\beta}(B)$ and every $i \in V_n$,

$$A_i \leq A_d.$$

Fifth, the exposed-order comparison between the block-energy scale and the binomial scale,

$$a d \binom{d}{k_\star} \leq d A_d \leq b d \binom{d}{k_\star}.$$

3. **Rescaling the frontier variable.** Since $n \geq 1 > 0$, dividing the exposed-order comparison of the preceding item by n gives

$$ax \leq y \leq bx.$$

The map $t \mapsto \min\{1, t\}$ is nondecreasing, so

$$\min\{1, ax\} \leq \min\{1, y\} \leq \min\{1, bx\}.$$

4. **Two constant comparisons.** First,

$$\min\{1, a^2\} \leq a.$$

Indeed, if $a \leq 1$ then $\min\{1, a^2\} \leq a^2 \leq a$, while if $a > 1$ then $\min\{1, a^2\} \leq 1 \leq a$. Second, since $0 < a \leq b$ we have $b > 0$, and

$$b \leq \max\{1, b^2\}.$$

Indeed, if $b \leq 1$ then $b \leq 1 \leq \max\{1, b^2\}$, while if $b > 1$ then $b \leq b^2 \leq \max\{1, b^2\}$.

5. **Lower rescaling.** We claim that

$$\min\{1, a^2\} \min\{1, x\} \leq a \min\{1, ax\}.$$

Split on the two saturations. If $x \leq 1$ and $ax \leq 1$, both minima are attained at their arguments and

$$\min\{1, a^2\} x \leq a^2 x = a(ax).$$

If $x \leq 1$ and $ax > 1$, then $\min\{1, x\} = x$, $\min\{1, ax\} = 1$, and, by the first comparison of the preceding item followed by $x \leq 1$,

$$\min\{1, a^2\} x \leq ax \leq a.$$

If $x > 1$ and $ax \leq 1$, then $\min\{1, x\} = 1$, $\min\{1, ax\} = ax$, and, using $x \geq 1$,

$$\min\{1, a^2\} \leq a^2 \leq a^2 x = a(ax).$$

If $x > 1$ and $ax > 1$, both minima equal 1 and the claim is again $\min\{1, a^2\} \leq a$.

6. **Upper rescaling.** We claim that

$$b \min\{1, bx\} \leq \max\{1, b^2\} \min\{1, x\}.$$

If $x \leq 1$ and $bx \leq 1$, then

$$b(bx) = b^2 x \leq \max\{1, b^2\} x.$$

If $x \leq 1$ and $bx > 1$, then $\min\{1, bx\} = 1$ and, using $bx \geq 1$ and $b > 0$,

$$b \leq b(bx) = b^2 x \leq \max\{1, b^2\} x.$$

If $x > 1$ and $bx \leq 1$, then $\min\{1, x\} = 1$ and $\min\{1, bx\} = bx$. Here $b < 1$: otherwise $b \geq 1$ would give $x \leq bx$, hence $1 < x \leq bx \leq 1$, a contradiction. Consequently

$$b(bx) \leq b \leq 1 \leq \max\{1, b^2\}.$$

If $x > 1$ and $bx > 1$, both minima equal 1 and the claim is the second comparison of the fourth item, $b \leq \max\{1, b^2\}$.

7. **The minimax lower bound.** Multiplying the lower rescaling of the fifth item by $B^2 \geq 0$, then inserting the min comparison of the third item, and finally the lower frontier of the second item,

$$\begin{aligned} c_{\beta,p} B^2 \min\{1, x\} &= B^2 (\min\{1, a^2\} \min\{1, x\}) \\ &\leq B^2 (a \min\{1, ax\}) \\ &\leq B^2 (a \min\{1, y\}) \\ &\leq R_n^*(d, \beta, p, B). \end{aligned}$$

This is the first assertion of the minimax frontier.

8. **The projected SNIPE estimator is an admissible competitor.** For every $(G, c) \in \mathcal{M}_{n,d,\beta}(B)$, every $i \in V_n$, and every $z \in \{0, 1\}^{V_n}$,

$$|Y_i(z)| = \left| \sum_{S \subseteq N_i} c_{i,S} \prod_{j \in S} z_j \right| \leq \sum_{S \subseteq N_i} |c_{i,S}| \leq B,$$

by the triangle inequality, the bound $|\prod_{j \in S} z_j| \leq 1$ valid because every factor lies in $\{0, 1\}$, and [assumption 4](#). Since $n \geq 1$, the population is nonempty and the target is the average of the unitwise all-treated versus all-control contrasts,

$$\tau_n = \frac{1}{n} \sum_{i \in V_n} (Y_i(\mathbf{1}) - Y_i(\mathbf{0})),$$

so

$$|\tau_n| \leq \frac{1}{n} \sum_{i \in V_n} (|Y_i(\mathbf{1})| + |Y_i(\mathbf{0})|) \leq 2B.$$

The estimator $\hat{\tau}_n^{\text{up}}$ of [definition 7](#) is the projection of $\hat{\tau}_n^{\text{SNIPE}}$ onto $[-B, B]$, which is a nonempty interval because $B \geq 0$; hence $|\hat{\tau}_n^{\text{up}}| \leq B$ pointwise in the assignment. Combining the last two bounds,

$$|\hat{\tau}_n^{\text{up}} - \tau_n| \leq |\hat{\tau}_n^{\text{up}}| + |\tau_n| \leq B + 2B = 3B,$$

so $(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \leq 9B^2$ pointwise, and averaging over the design,

$$\mathbb{E}[(\hat{\tau}_n^{\text{up}} - \tau_n)^2] \leq 9B^2 \quad \text{for every } (G, c) \in \mathcal{M}_{n,d,\beta}(B).$$

Thus the modelwise risks of $\hat{\tau}_n^{\text{up}}$ are bounded above by a single constant; the estimator is also measurable in the observed outcomes for each fixed graph and assignment, being a clipped linear combination of them. So $\hat{\tau}_n^{\text{up}}$ is one of the estimators over which [definition 8](#) takes its infimum.

Moreover $\mathcal{M}_{n,d,\beta}(B)$ is nonempty: the empty graph has all neighborhoods empty, hence both degrees $0 \leq d$, and the zero coefficient schedule is supported on those neighborhoods, has no monomial of order above β , and has coefficient mass $0 \leq B$. Every competitor's worst-case risk is therefore a supremum of design mean squared errors over a nonempty class, so it is at least 0, and the set of worst-case risks appearing in [definition 8](#) is nonempty and bounded below. Its infimum is at most the value attained at the admissible competitor $\hat{\tau}_n^{\text{up}}$:

$$R_n^*(d, \beta, p, B) \leq \sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}[(\hat{\tau}_n^{\text{up}} - \tau_n)^2].$$

9. **The minimax upper bound.** Starting from the projected-SNIPE upper bound of the second item, then inserting the min comparison of the third item and the upper rescaling of the sixth item, each multiplied by $B^2 \geq 0$,

$$\begin{aligned} \sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] &\leq bB^2 \min\{1, y\} \\ &\leq B^2 (b \min\{1, bx\}) \\ &\leq B^2 (\max\{1, b^2\} \min\{1, x\}) = C_{\beta,p} B^2 \min\{1, x\}. \end{aligned}$$

Together with the preceding two items, this completes the minimax frontier assertion.

10. **The unprojected SNIPE estimator.** From the unprojected-SNIPE bound of the second item, then $y \leq bx$, then $b \leq \max\{1, b^2\}$ together with $x \geq 0$,

$$\begin{aligned} \sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] &\leq B^2 y \\ &\leq B^2 (bx) \\ &\leq C_{\beta,p} B^2 x. \end{aligned}$$

11. **Score energy.** The score-energy assertion $A_i \leq A_d$ is the local-energy comparison recorded as the fourth clause of the second item.
12. **Overlap counts.** Let $(G, c) \in \mathcal{M}_{n,d,\beta}(B)$, let $i \in V_n$, and let $r \geq 1$. By [definition 5](#) the graph G satisfies [assumption 2](#) with bound d , so [lemma 2](#), applied to G , the degree bound d , the order r , and the unit i , gives precisely the asserted identity and the two asserted bounds.
13. **Block construction.** Assume $B > 0$ and $d \geq 1$. In the asserted clause, m , ρ , and δ are the quantities of [definition 10](#): $m = \lfloor n/d \rfloor$ is the number of complete d -blocks,

$$\rho = \frac{md}{n}$$

is the active population fraction, and

$$\delta = B \min\{(2H_{\beta,p})^{-1}, (4\pi)^{-1}\} \min\{1, \sqrt{A_d/m}\}$$

is the least-favourable perturbation amplitude, where

$$H_{\beta,p} = \sup_{d \geq 1} \sum_T |h_{d,T}|$$

is the representer mass bound of [definition 10](#) and $h_{d,T}$ are the raw-monomial coefficients of the normalized block representer of [definition 9](#). The two schedules attached to a baseline vector $U = (U_1, \dots, U_m)$ with $|U_b| \leq B/2$ for all b are the sign- σ schedules

$$c_{i,T} = U_b \mathbf{1}\{T = \emptyset\} + \sigma \delta h_{d,T}, \quad |T| \leq k_*,$$

of [definition 10](#) with $\sigma = +1$ and $\sigma = -1$, all remaining coefficients being zero, carried by the complete d -block graph; Π_+ and Π_- are the corresponding priors of [definitions 11](#)

and 12. With these readings, the membership of M_+ and M_- in $\mathcal{M}_{n,d,\beta}(B)$, the two target values

$$\tau_n(M_+) = \rho\delta, \quad \tau_n(M_-) = -\rho\delta,$$

the squared Hellinger bound

$$H^2(\Pi_+, \Pi_-) \leq \frac{4\pi^2 m \delta^2}{B^2 A_d},$$

the bracketing $\frac{1}{2} \leq \rho \leq 1$, and the identity

$$\frac{A_d}{m} = \frac{dA_d/n}{\rho}$$

are verbatim the least-favourable block clause of [theorem 3](#) at the same n , d , and B . This proves the block-construction assertion and completes the proof. $\square$

Proof of [theorem 3](#). 1. **The two constants.** By [lemma 1](#) there are constants c_1, c_2, K , depending only on (β, p) , with $0 < c_1 \leq c_2$ and $0 < K$, such that for every $d \geq 1$

$$c_1 \binom{d}{k_\star(d, \beta, p)} \leq A_d \leq c_2 \binom{d}{k_\star(d, \beta, p)}, \quad \sum_{T \subseteq \{1, \dots, d\}} |h_{d,T}| \leq K.$$

In particular the quantity $H_{\beta,p} = \sup_{d \geq 1} \sum_T |h_{d,T}|$ of [definition 10](#) is finite, with $H_{\beta,p} \leq K$. It is also at least one: evaluating the raw expansion of [definition 9](#) at the all-treated and the all-control assignment of a single-coordinate block and subtracting gives

$$1 = L_1(h_1) = \sum_{\emptyset \neq T \subseteq \{1\}} h_{1,T} \leq \sum_{T \subseteq \{1\}} |h_{1,T}| \leq H_{\beta,p},$$

where $L_1(h_1) = 1$ is the representer identity of [lemma 1](#). Now set

$$\kappa := \min\{(2H_{\beta,p})^{-1}, (4\pi)^{-1}\}, \quad c_{\beta,p} := \min\left\{c_1, \frac{\kappa^2}{16}\right\}, \quad C_{\beta,p} := \max\{16, c_2\}.$$

Since $H_{\beta,p} \geq 1 > 0$ and $\pi > 0$, both entries in the definition of κ are positive, so $\kappa > 0$ and therefore $c_{\beta,p} > 0$. Also

$$c_{\beta,p} \leq c_1 \leq c_2 \leq C_{\beta,p}.$$

With this κ , the perturbation amplitude of [definition 10](#) reads

$$\delta = B\kappa\eta, \quad \eta := \min\left\{1, \sqrt{A_d/m}\right\}.$$

2. **Standing data.** Fix $n \geq 1$, a design D satisfying [assumption 1](#) with probability p , an integer d with $d \leq n$, and $B \geq 0$. Since $p(1-p) > 0$, every summand of A_d in [definition 9](#) is nonnegative, so $A_d \geq 0$; and by [definition 2](#) $A_0 = 0$. For $d \geq 1$ the order $r = 1$ contributes $\binom{d}{1} \Delta_1(p)^2 / [p(1-p)] = d / [p(1-p)] > 0$ to A_d , so

$$0 < A_d \quad (d \geq 1).$$

3. **Model comparison.** Let $(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)$. Every raw monomial takes values in $[0, 1]$, so for every unit i and every assignment z

$$|Y_i(z)| = \left| \sum_{S \subseteq N_i} c_{i,S} \prod_{j \in S} z_j \right| \leq \sum_{S \subseteq N_i} |c_{i,S}| \leq B,$$

the last step being [assumption 4](#). The remaining requirements of [definition 20](#) — neighborhood support and [assumption 3](#) — are already part of [definition 4](#), and the graph requirement is unchanged. Hence the same pair (G_n, c) lies in $\mathcal{M}_{n,d,\beta}^\infty(B)$: the containment holds carrier by carrier.

Now let $B > 0$ and $d \geq 1$. Pick a unit $i_0 \in V_n$ (possible since $n \geq 1$), let G^0 consist of the single loop $i_0 \rightarrow i_0$, and set

$$c_{i_0, \emptyset}^0 := B, \quad c_{i_0, \{i_0\}}^0 := -2B,$$

with all other coefficients zero. Then $N_{i_0} = \{i_0\}$ and $N_i = \emptyset$ for $i \neq i_0$, so both degrees in [assumption 2](#) are at most $1 \leq d$, and the only nonzero coefficients are indexed by sets of cardinality at most $1 \leq \beta$. The induced outcomes are

$$Y_{i_0}(z) = B - 2Bz_{i_0} \in \{B, -B\}, \quad Y_i(z) = 0 \quad (i \neq i_0),$$

so $(G^0, c^0) \in \mathcal{M}_{n,d,\beta}^\infty(B)$. Its coefficient mass at i_0 is

$$\sum_{S \subseteq N_{i_0}} |c_{i_0, S}^0| = B + 2B = 3B > B,$$

so no member of $\mathcal{M}_{n,d,\beta}(B)$ has graph G^0 and schedule c^0 , and the containment is strict.

4. **Local energy.** Let (G_n, c) belong to either model class and let $i \in V_n$. By [assumption 2](#), $|N_i| \leq d$, hence $\min\{\beta, |N_i|\} \leq \bar{\beta}_d$ and $\binom{|N_i|}{r} \leq \binom{d}{r}$ for every r . Since each term $\Delta_r(p)^2/[p(1-p)]^r$ is nonnegative,

$$A_i = \sum_{r=1}^{\min\{\beta, |N_i|\}} \binom{|N_i|}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r} \leq \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r} = A_d.$$

The estimate uses only the degree bound of the carrier graph, so it holds verbatim on $\mathcal{M}_{n,d,\beta}(B)$ and on $\mathcal{M}_{n,d,\beta}^\infty(B)$.

5. **Unbiasedness of SNIPE.** Under [assumption 1](#) the design D is the common- p product Bernoulli law, so the centered monomials in [definition 6](#) obey, for every unit i and every nonempty $S \subseteq N_i$ with $|S| \leq \min\{\beta, |N_i|\}$,

$$\mathbb{E}_Z \left[g_{i,\beta,p}(Z) \prod_{j \in S} Z_j \right] = 1, \quad \mathbb{E}_Z [g_{i,\beta,p}(Z)] = 0.$$

The first identity collapses the double sum in $g_{i,\beta,p}$ to the binomial contrast $\sum_{r=1}^{|S|} \binom{|S|}{r} \Delta_r(p) p^{|S|-r} = 1$; the second is the vanishing design mean of every non-constant centered monomial. Expanding Y_i over its raw monomials, using [assumption 3](#) to discard the terms with $|S| > \beta$, and applying the two identities term by term gives

$$\mathbb{E}_Z [g_{i,\beta,p}(Z) Y_i(Z)] = \sum_{\emptyset \neq S \subseteq N_i} c_{i,S} = Y_i(\mathbf{1}) - Y_i(\mathbf{0}).$$

Averaging over $i \in V_n$ yields $\mathbb{E}_Z[\hat{\tau}_n^{\text{SNIPE}}] = \tau_n$ at every model of $\mathcal{M}_{n,d,\beta}(B)$ and of $\mathcal{M}_{n,d,\beta}^\infty(B)$, since only the low-order and support properties of the schedule were used.

6. **Unclipped SNIPE risk.** Fix a model in either class and write

$$W_i(z) := Y_i(z)g_{i,\beta,p}(z), \quad F_i(z) := W_i(z) - \mathbb{E}_Z[W_i(Z)].$$

Each F_i has design mean zero and depends on the assignment only through $\{Z_j : j \in N_i\}$, because both Y_i and $g_{i,\beta,p}$ do. Expanding each F_i in the orthogonal basis of centered monomials $\prod_{j \in S} (Z_j - p)$, only subsets $S \subseteq N_i$ receive a nonzero coefficient, and the empty subset receives none. For a fixed nonempty S , the units that can contribute are those with $S \subseteq N_i$, and there are at most d of them: choosing $j \in S$ places every such unit in the out-neighborhood of j , which is the out-degree count in [lemma 2](#). Applying Cauchy–Schwarz to each such group of at most d coefficients and summing over S gives

$$\mathbb{E}_Z \left[\left(\sum_{i \in V_n} F_i(Z) \right)^2 \right] \leq d \sum_{i \in V_n} \mathbb{E}_Z [F_i(Z)^2].$$

For each unit, $|Y_i(z)| \leq B$ — by the display of item 3 on $\mathcal{M}_{n,d,\beta}(B)$, and by definition on $\mathcal{M}_{n,d,\beta}^\infty(B)$ — while the exact score energy is $\mathbb{E}_Z[g_{i,\beta,p}(Z)^2] = A_i$. Hence, using item 4,

$$\mathbb{E}_Z [F_i(Z)^2] \leq \mathbb{E}_Z [W_i(Z)^2] \leq B^2 \mathbb{E}_Z [g_{i,\beta,p}(Z)^2] = B^2 A_i \leq B^2 A_d.$$

By item 5 the estimator is unbiased, so its mean squared error equals its variance, and

$$\mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{1}{n^2} \mathbb{E}_Z \left[\left(\sum_{i \in V_n} F_i(Z) \right)^2 \right] \leq \frac{d}{n^2} \cdot n \cdot B^2 A_d = B^2 \frac{d A_d}{n}.$$

Taking the supremum over each class,

$$\sup_{\mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] \leq B^2 \frac{d A_d}{n}, \quad \sup_{\mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] \leq B^2 \frac{d A_d}{n}.$$

7. **Target envelopes and clipping.** On $\mathcal{M}_{n,d,\beta}(B)$ the all-treated versus all-control contrast of a single unit is $Y_i(\mathbf{1}) - Y_i(\mathbf{0}) = \sum_{\emptyset \neq S \subseteq N_i} c_{i,S}$, so by [assumption 4](#)

$$|\tau_n| \leq \frac{1}{n} \sum_{i \in V_n} \sum_{\emptyset \neq S \subseteq N_i} |c_{i,S}| \leq B,$$

whereas on $\mathcal{M}_{n,d,\beta}^\infty(B)$ the uniform outcome envelope gives

$$|\tau_n| \leq \frac{1}{n} \sum_{i \in V_n} (|Y_i(\mathbf{1})| + |Y_i(\mathbf{0})|) \leq 2B.$$

For $R \geq 0$ and any target t with $|t| \leq R$, Euclidean projection onto $[-R, R]$ is a contraction towards t :

$$(\Pi_{[-R,R]}(x) - t)^2 \leq (x - t)^2 \quad \text{for every } x \in \mathbb{R}.$$

Applying this with $R = B$, $t = \tau_n$ on $\mathcal{M}_{n,d,\beta}(B)$, and with $R = 2B$, $t = \tau_n$ on $\mathcal{M}_{n,d,\beta}^\infty(B)$, gives modelwise

$$\mathbb{E}_Z \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] \leq \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right], \quad \mathbb{E}_Z \left[(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2 \right] \leq \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right].$$

The projections also give saturated bounds that do not involve the design at all: $|\hat{\tau}_n^{\text{up}}| \leq B$ together with $|\tau_n| \leq B$ yields $(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \leq 4B^2$ pointwise, and $|\hat{\tau}_{n,\infty}^{\text{up}}| \leq 2B$ together with $|\tau_n| \leq 2B$ yields $(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2 \leq 16B^2$ pointwise. Using item 6 when $dA_d/n \leq 1$ and the saturated bounds when $dA_d/n > 1$,

$$\sup_{\mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] \leq 4B^2 \min \left\{ 1, \frac{dA_d}{n} \right\},$$

$$\sup_{\mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2 \right] \leq 16B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

Since $4 \leq 16 \leq C_{\beta,p}$ and $B^2 \min\{1, dA_d/n\} \geq 0$, both right-hand sides are at most $C_{\beta,p} B^2 \min\{1, dA_d/n\}$, which proves the coefficient-class clipped bound of the theorem.

8. Ordering of the two minimax risks. The identity

$$R_{n,\ell_1}^*(d, \beta, p, B) = R_n^*(d, \beta, p, B)$$

is the definition of the coefficient-mass risk in [definition 22](#).

By item 3, each $(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)$ is also a member of $\mathcal{M}_{n,d,\beta}^\infty(B)$ with the same graph and the same schedule, hence with the same observed outcomes and the same target; the modelwise risk of any estimator is therefore unchanged by the embedding. Consequently, for every graph-aware estimator $\hat{\tau}_n$,

$$\sup_{\mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right] \leq \sup_{\mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right],$$

and an estimator whose bounded-outcome risk range is bounded above has bounded coefficient-mass risk range as well, so it competes in the infimum defining R_{n,ℓ_1}^* . Taking the infimum over the estimators admissible for the bounded-outcome problem gives

$$R_{n,\ell_1}^*(d, \beta, p, B) \leq R_{n,\infty}^*(d, \beta, p, B).$$

The estimator $\hat{\tau}_{n,\infty}^{\text{up}}$ of [definition 23](#) is measurable in the observed outcomes, and by the saturated bound of item 7 its modelwise risk over $\mathcal{M}_{n,d,\beta}^\infty(B)$ never exceeds $16B^2$, so its risk range is bounded above and it is admissible in [definition 22](#). Hence the infimum is at most its worst-case risk, and with item 7,

$$R_{n,\infty}^*(d, \beta, p, B) \leq \sup_{\mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2 \right] \leq C_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

9. The least-favourable block family. Assume $B > 0$ and $d \geq 1$, and use the notation of [definition 10](#): $m = \lfloor n/d \rfloor$, $q = md$, $\rho = q/n$, $s = B/2$, and $\delta = B\kappa\eta$ as in item 1.

Since $1 \leq d \leq n$ we have $m \geq 1$. Fix $U = (U_1, \dots, U_m)$ with $|U_b| \leq B/2$ for all b , and let $\sigma \in \{-1, +1\}$.

Membership. The signed schedule of [definition 10](#) assigns to each active unit $i \in V_b$ the coefficients $c_{i,T} = U_b \mathbf{1}\{T = \emptyset\} + \sigma \delta h_{d,T}$ for $T \subseteq V_b$, and zero to every other coefficient. It is supported in $N_i = V_b$ on the complete-block graph, and it vanishes for $|T| > \beta$ because $h_{d,T} = 0$ there, so [assumption 3](#) holds; the complete-block graph has both degrees equal to d , so [assumption 2](#) holds. For the mass, $\eta \leq 1$ and $\kappa \leq (2H_{\beta,p})^{-1}$ give

$$\sum_{T \subseteq N_i} |c_{i,T}| \leq |U_b| + |\delta| \sum_T |h_{d,T}| \leq \frac{B}{2} + B\kappa H_{\beta,p} \leq \frac{B}{2} + \frac{B}{2} = B.$$

Hence both signed schedules define models M_+ and M_- in $\mathcal{M}_{n,d,\beta}(B)$ carried by the complete-block graph.

Targets. On an active unit the baseline U_b cancels in the all-treated versus all-control contrast, and $L_d(h_d) = h_d(\mathbf{1}_d) - h_d(\mathbf{0}_d) = 1$ by [lemma 1](#), so

$$Y_i(\mathbf{1}) - Y_i(\mathbf{0}) = \sigma \delta \quad (i \text{ active}), \quad Y_i(\mathbf{1}) - Y_i(\mathbf{0}) = 0 \quad (i \text{ inactive}).$$

There are $q = md$ active units, so

$$\tau_n(M_\sigma) = \frac{q}{n} \sigma \delta = \sigma \rho \delta, \quad \text{i.e.} \quad \tau_n(M_+) = \rho \delta, \quad \tau_n(M_-) = -\rho \delta.$$

Hellinger distance. Every unit of an active block shares the block's outcome value, so the observation reduces to the pair $(Z, (Y_b)_{b \leq m})$, whose law under Π_σ has density

$$\Pr(Z = z) \prod_{b=1}^m f_s(y_b - \sigma \delta h_d(z_{V_b}))$$

against counting measure on assignments times m -dimensional Lebesgue measure. Conditionally on $Z = z$, the two laws Π_+ and Π_- are products of m translates of f_s with separations $2\delta h_d(z_{V_b})$, and the cosine-squared density has quadratic affinity defect

$$1 - \int \sqrt{f_s(u - a)f_s(u - a')} du \leq \frac{\pi^2(a - a')^2}{8s^2}.$$

Summing the defect over the m coordinates and using $H^2 = 2(1 - \text{affinity})$ with $s = B/2$ gives the conditional bound

$$H^2(\Pi_+(\cdot | z), \Pi_-(\cdot | z)) \leq \frac{4\pi^2\delta^2}{B^2} \sum_{b=1}^m h_d(z_{V_b})^2.$$

Because the assignment marginal is common to the two laws, the unconditional squared Hellinger distance is the design average of the conditional ones. Each block sees an independent copy of the Bernoulli law, so $\mathbb{E}_Z[h_d(Z_{V_b})^2] = A_d^{-1}$ by [lemma 1](#), whence $\mathbb{E}_Z[\sum_b h_d(Z_{V_b})^2] = m/A_d$ and

$$H^2(\Pi_+, \Pi_-) \leq \frac{4\pi^2 m \delta^2}{B^2 A_d}.$$

Active share. From $q = md \leq n$ we get $\rho \leq 1$. Writing $n = q + (n - q)$ with $n - q < d \leq q$ (the second inequality because $m \geq 1$) gives $n < 2q$, hence

$$\frac{1}{2} \leq \rho \leq 1.$$

Finally, $q = md$ and $\rho = q/n$ give $\rho A_d/m = dA_d/n$, that is,

$$\frac{A_d}{m} = \frac{dA_d/n}{\rho}.$$

10. **Lower frontier.** If $d = 0$ then $A_0 = 0$ by [definition 2](#) and the left side of the claimed bound is 0; if $B = 0$ it is 0 as well. In both cases the bound holds because the minimax risk is nonnegative: every modelwise risk is a mean squared error, the zero estimator is admissible so the infimum is over a nonempty set, and the empty graph with the zero schedule is a member of $\mathcal{M}_{n,d,\beta}(B)$.

Assume now $B > 0$ and $d \geq 1$, and keep the notation of item 9. Since $\eta \leq \sqrt{A_d/m}$ we have $\eta^2 \leq A_d/m$, i.e.

$$m\eta^2 \leq A_d,$$

and since $\kappa \leq (4\pi)^{-1}$ we have $16\pi^2\kappa^2 \leq 1$, i.e. $4\pi^2\kappa^2 \leq \frac{1}{4}$. Substituting $\delta = B\kappa\eta$ into the Hellinger bound of item 9,

$$H^2(\Pi_+, \Pi_-) \leq \frac{4\pi^2 m \delta^2}{B^2 A_d} = 4\pi^2 \kappa^2 \frac{m\eta^2}{A_d} \leq \frac{1}{4} \cdot \frac{m\eta^2}{A_d} \leq \frac{1}{4}.$$

In the unhalved convention the total variation distance is bounded by the Hellinger distance, so

$$\sup_A |\Pi_+(A) - \Pi_-(A)| \leq \sqrt{H^2(\Pi_+, \Pi_-)} \leq \frac{1}{2}.$$

The two targets $\pm\rho\delta$ of item 9 are $2\rho\delta$ apart. Let $\hat{\tau}_n$ be any admissible estimator and let T be its value read on the block construction, a measurable function of $(Z, (Y_b)_{b \leq m})$. Le Cam's two-point bound gives

$$\max \left\{ \Pi_+(|T - \rho\delta| \geq \rho\delta), \Pi_-(|T + \rho\delta| \geq \rho\delta) \right\} \geq \frac{1 - \frac{1}{2}}{2} = \frac{1}{4}.$$

Each of the two probabilities is dominated by the worst-case risk. Indeed, for each sign σ and each baseline vector U in the support of the prior, the schedule of item 9 is a member of $\mathcal{M}_{n,d,\beta}(B)$ whose target is $\sigma\rho\delta$, and pointwise in the assignment

$$(\rho\delta)^2 \mathbf{1}\{|\hat{\tau}_n - \sigma\rho\delta| \geq \rho\delta\} \leq (\hat{\tau}_n - \sigma\rho\delta)^2,$$

so taking design expectations and then averaging over U — a probability law — gives

$$(\rho\delta)^2 \Pi_\sigma(|T - \sigma\rho\delta| \geq \rho\delta) \leq \sup_{\mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right].$$

Combining the last two displays, every admissible estimator has worst-case risk at least $(\rho\delta)^2/4$, and taking the infimum over admissible estimators,

$$\frac{(\rho\delta)^2}{4} \leq R_{n,\ell_1}^*(d, \beta, p, B).$$

It remains to convert the scale. By the identity of item 9 and $\rho \leq 1$,

$$\frac{dA_d}{n} = \rho \frac{A_d}{m} \leq \frac{A_d}{m}, \quad \text{hence} \quad \min\left\{1, \frac{dA_d}{n}\right\} \leq \min\left\{1, \frac{A_d}{m}\right\} = \eta^2,$$

the last equality because $\eta = \min\{1, \sqrt{A_d/m}\}$. Also $\rho \geq \frac{1}{2}$ gives $\rho^2/4 \geq 1/16$. Therefore, using $c_{\beta,p} \leq \kappa^2/16$,

$$\begin{aligned} c_{\beta,p} B^2 \min\left\{1, \frac{dA_d}{n}\right\} &\leq \frac{\kappa^2}{16} B^2 \eta^2 \\ &\leq \frac{\rho^2}{4} B^2 \kappa^2 \eta^2 \\ &= \frac{(\rho B \kappa \eta)^2}{4} = \frac{(\rho \delta)^2}{4} \leq R_{n,\ell_1}^*(d, \beta, p, B). \end{aligned}$$

11. **Binomial scale.** Let $d \geq 1$ and write $k_\star = k_\star(d, \beta, p)$. Multiplying the energy scale of [lemma 1](#) by $d \geq 0$ and using $c_{\beta,p} \leq c_1$ and $c_2 \leq C_{\beta,p}$ from item 1,

$$c_{\beta,p} d \binom{d}{k_\star} \leq c_1 d \binom{d}{k_\star} \leq dA_d \leq c_2 d \binom{d}{k_\star} \leq C_{\beta,p} d \binom{d}{k_\star}.$$

For $d = 0$ all three quantities vanish, since the factor d is zero.

12. **Exact complete-block risk.** Assume $d \geq 1$ and $d \mid n$, put $m = n/d$, and let G be the disjoint union of m complete directed d -blocks with loops. Since $n = md$,

$$\frac{B^2 A_d}{m} = B^2 \frac{dA_d}{n}.$$

Suppose first $B > 0$. On G every unit has $N_i = V_{b(i)}$ with $|N_i| = d$, so the SNIPE score of [definition 6](#) is exactly the complete-block score of [definition 9](#) read on that unit's block,

$$g_{i,\beta,p}(z) = g_d(z_{V_{b(i)}}),$$

and consequently $\hat{\tau}_n^{\text{SNIPE}} = \hat{\tau}_{w,t}$ for the block-local weights $w_{i,t} := g_d$, which satisfy the mean-zero and raw-moment restrictions of [definition 18](#) by [lemma 1](#). Apply [theorem 2](#) at the index with $n_t = n$, $d_t = d$, $m_t = m$ and $G_t = G$. It gives the exact blockwise representation of the fixed-graph worst-case risk and the lower bound $d^2 A_d \leq \Psi_{b,t}(w)$ for every admissible w . For the canonical choice $w_{i,t} = g_d$ the matching upper bound

$$\Psi_{b,t}(w) \leq d^2 A_d$$

holds on every block V_b . Indeed, fix a block V_b and a signed choice $(\varepsilon_i, S_i)_{i \in V_b}$ admissible in the maximum defining $\Psi_{b,t}$. The d summands are not copies of one another — they carry different subsets S_i and different signs ε_i — but every unit of the block carries the same weight $g_d(Z_b)$, which therefore factors out:

$$\sum_{i \in V_b} \varepsilon_i \{g_d(Z_b) Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\} = g_d(Z_b) \Phi(Z_b) - \Lambda,$$

$$\Phi(Z_b) := \sum_{i \in V_b} \varepsilon_i Z_{S_i}, \quad \Lambda := \sum_{i \in V_b} \varepsilon_i \mathbf{1}(S_i \neq \emptyset).$$

The representer identities of [lemma 1](#) give $\mathbb{E}_Z[g_d(Z_b)Z_S] = 1$ for every nonempty $S \subseteq V_b$ with $|S| \leq \beta_d$ and $\mathbb{E}_Z[g_d(Z_b)] = 0$, so $\mathbb{E}_Z[g_d(Z_b)\Phi(Z_b)] = \Lambda$ and expanding the square removes the centering term,

$$\mathbb{E}_Z \left[(g_d(Z_b)\Phi(Z_b) - \Lambda)^2 \right] = \mathbb{E}_Z \left[(g_d(Z_b)\Phi(Z_b))^2 \right] - \Lambda^2 \leq \mathbb{E}_Z \left[(g_d(Z_b)\Phi(Z_b))^2 \right].$$

Each of the d terms of Φ has modulus at most one, since $0 \leq Z_{S_i} \leq 1$ and $|\varepsilon_i| = 1$, so $|\Phi(Z_b)| \leq d$ pointwise and therefore

$$\mathbb{E}_Z \left[(g_d(Z_b)\Phi(Z_b))^2 \right] \leq d^2 \mathbb{E}_Z [g_d(Z_b)^2] = d^2 A_d,$$

the last equality being the exact block-score energy $\mathbb{E}_Z[g_d(Z)^2] = A_d$ of [lemma 1](#). Maximizing over admissible choices gives the displayed upper bound. Hence $\Psi_{b,t}(w) = d^2 A_d$ on every block, and

$$\sup_{c \in \mathcal{M}_t(G)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{B^2}{n^2} \sum_{b=1}^m \Psi_{b,t}(w) = \frac{B^2 m d^2 A_d}{n^2} = \frac{B^2 A_d}{m}.$$

If instead $B = 0$, the only admissible schedule on any graph has zero coefficient mass, so $\tau_n = 0$, while item 6 bounds every modelwise SNIPE risk by $B^2 d A_d / n = 0$; as design mean squared errors are nonnegative, the fixed-graph coefficient-class risk again equals $0 = B^2 A_d / m$.

For all $B \geq 0$ the four risks now coincide. Restricting the supremum to a single graph can only decrease it, so the fixed-graph coefficient-class risk is at most the global coefficient-class risk, and the fixed-graph bounded-outcome risk is at most the global bounded-outcome risk; by item 3 each coefficient-mass model on G embeds in the bounded-outcome class without changing the graph, the schedule or the risk, so the fixed-graph coefficient-class risk is at most the fixed-graph bounded-outcome risk. Item 6 bounds both global risks by $B^2 d A_d / n = B^2 A_d / m$, which is the value just computed for the fixed-graph coefficient-class risk. The chain of inequalities therefore closes, and

$$\begin{aligned} \sup_{c \in \mathcal{M}_t(G)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] &= \sup_{\substack{(G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B) \\ G_n = G}} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \sup_{\mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] \\ &= \sup_{\mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{B^2 A_d}{m} = B^2 \frac{d A_d}{n}. \end{aligned}$$

Item 3 is the model comparison; items 5 and 6 are the unbiasedness and the worst-case bound $B^2 d A_d / n$ for unclipped SNIPE; items 7, 8 and 10 are the frontier bounds; items 4 and 11 are the local energy and the binomial scale; item 12 is the exact block risk; and item 9 is the least-favourable block family. Together these are the six assertions of the theorem, with the constants $c_{\beta,p}$ and $C_{\beta,p}$ fixed in item 1 independently of n , D , d and B . $\square$

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